

Towards Scale-up of Graphene Production via Nonoxidizing Liquid Exfoliation Methods

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Graphene, the two-dimensional form of carbon, has received a great deal of attention across academia and industry due to its extraordinary electrical, mechanical, thermal, chemical, and optical properties. In view of the potential impact of graphene on numerous and diverse applications in electronics, novel materials, energy, transport, and healthcare, large-scale graphene production is a challenge that must be addressed. In the past decade, top-down production has demonstrated high potential for scale-up. This review features the recent progress made in top-down production methods that have been proposed for the manufacturing of graphene-based products. Fabrication methods such as liquid-phase mechanical, chemical and electrochemical exfoliation of graphite are outlined, with a particular focus on nonoxidizing routes for graphene production. Analysis of exfoliation mechanisms, solvent considerations, key advantages and issues, and important production characteristics including production rate and yield, where applicable, are outlined. Future challenges and opportunities in graphene production are also highlighted. © 2018 American Institute of Chemical Engineers AIChE J, 64: 3246–3276, 2018

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Introduction

Nanomaterials have emerged as a critical sector that is likely to push the frontiers of technology. The remarkable properties which carbon nanomaterials boast, such as high electrical and thermal conductivities and tensile strength, offer a strong platform to exceed the current boundaries imposed in several fields of science and engineering. Within these fields, the application of fundamental chemical engineering concepts is crucial to understanding low-dimensional nanomaterials.¹ Two-dimensional (2-D) materials are one type of nanomaterial which is receiving considerable attention, the most featured one being undoubtedly graphene, a semimetal that consists of a flat monolayer of carbon atoms.² This carbon monolayer is densely packed into a 2-D honeycomb lattice and has fully conjugated sp^2 hybridized planar structure with carbon-carbon bond length of 0.142 nm. Due to its two-dimensional nature, graphene has a unique range of properties and is part of a family of carbon-based materials with other dimensionalities, for example, 3-D graphite, 1-D nanotubes, and 0-D fullerenes.

Scientific research on graphene over the past decade has demonstrated a remarkable range of properties. The electron mobility at room temperature can reach $> 250,000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.³ It can deliver a maximum current density that is more than several million times higher than that of copper. Thermal conductivity and mechanical properties have also shown astonishing results. It has a thermal conductivity of approximately $3000 \text{ W m}^{-1} \text{ K}^{-1}$,⁴ a Young's modulus of 0.5–1.0 TPa and a high intrinsic strength of approximately 130 GPa.⁵ Moreover, its transmittance reaches

about 97.7%⁶ and it has extremely high resistance to gas permeation.⁷ Given the extraordinary breadth of high performing properties from this single material, and the recent emergence of other 2-D materials such as boron nitride nanosheets (BNNS), transition metal dichalcogenide monolayers (TMDs), and transition metal oxides (TMOs) also showing exceptional characteristics,^{8,9} this area of nanomaterials can play a major role in the future development of applications that will have tremendous economic and societal impact; these include optoelectronics, sensors, tissue engineering, drug delivery, energy conversion, and storage.

Despite the extensive list of potential end-uses, the implementation of graphene and other 2-D materials into practical applications is currently limited. There is widespread consensus within the scientific community that this is primarily attributed to the limitations of large-scale production strategies. A review by Ferrari et al.¹⁰ on the scientific and technological developments of graphene concluded that widespread practical implementation will be enabled once each production route is at a sufficiently mature stage.

In general, there are two ways to categorize the production of graphene: “bottom-up,” and “top-down.” Bottom-up techniques involve the utilisation of chemical reactions to produce 2-D graphene from hydrocarbon precursors, and include processes such as chemical vapour deposition, arc discharge, and epitaxial growth. The top-down approach uses graphite as a precursor, which is a crystalline material that is essentially constructed from multiple graphene sheets with an interplanar spacing of 0.335 nm. By exceeding the critical exfoliation energy of approximately 2 eV nm^{-2} ,¹¹ van der Waals forces between the graphene sheets are overcome, thereby deconstructing graphite into sheets of graphene. These sheets must then be held in an environment that minimizes the potential

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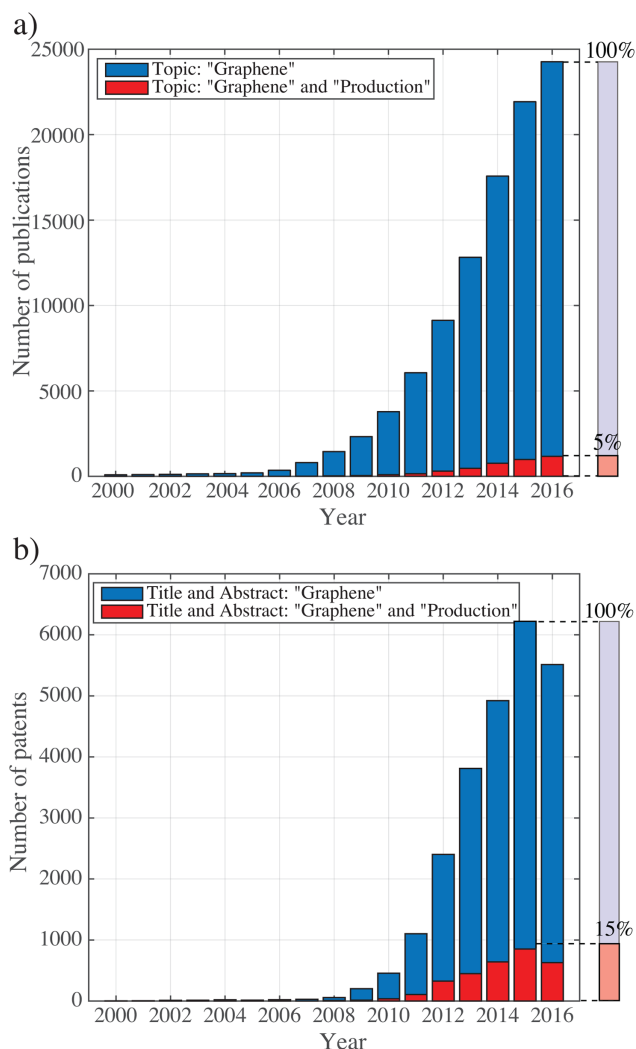


Figure 1. Publications and Patents in the field of graphene and graphene production..

a) Number of publications on Graphene and Graphene Production from 2000 to 2016. This information was sourced from ISI Web of science. Search criteria for both are: Topic = "Graphene" and Topic = "Graphene" and "Production" respectively. b) Number of patents on Graphene and Graphene production from 2000 to 2016. This information was sourced through a worldwide search using the European Patent Offices Espacenet database. Search criteria for both are: Title and Abstract = "Graphene" and Title and Abstract = "Graphene" and "Production," respectively. Data for 2016 is incomplete and lower than expected as there is a typical timeline between filing and publication [Color figure can be viewed at wileyonlinelibrary.com]

for restacking, commonly achieved using appropriate solvents or aqueous surfactant solutions. Top-down approaches, that can produce graphene in this liquid carrier fluid, also indirectly benefit certain end-use applications such as the deposition of graphene films for fabrication of photonic devices with both transparent and conducting properties.¹²

An obvious route towards reducing the cost of mass-production is through the development of methods that can deliver high yield and production rates. The material properties are also sensitive to the approach,¹³ which makes the production challenges complex and multifaceted. Many of the contemporary bottom-up and top-down production

approaches have evolved over the last ten years. The aim of the current review, therefore, is to provide an overview of the present status of large-scale graphene production. Particular emphasis is placed on the top-down route, highlighting the mechanisms, dispersion, advantages, and limitations of the existing techniques. Nonoxidizing liquid-phase exfoliation has shown potential for scaling-up the production of high quality graphene within this domain.

Publications and Patents

The emerging area of graphene production is receiving extensive attention, and a continuation of these research efforts can only benefit the future implementation of efficient production technologies. Within the broader context of graphene research activities, however, investigations on production are a fraction of the total. This is illustrated in Figure 1 by the number of publications and patents on these topics since 2000. As of 2016, publications under the topic of graphene production account for less than 5% of the total. Although it is a relatively low contribution (It is worth noting that the search terms "graphene" and "production" do not capture all studies on graphene production), the outlook appears promising when the yearly growth is examined from this historical information.

The trends in research activities over the 2000–2016 duration are presented in Figure 2. For both graphene and graphene production, there are two trends which describe a before and after state when research efforts increased considerably. For graphene research, there is no doubt that the breakthrough by Novoselov et al.² changed the broader graphene research path. The scientific output increased two orders of magnitude over a decade from this point in time. Figure 2 also shows that an equivalent point in time, where attention on the production of graphene began to increase, arrived about two years later. Since 2007, the rate of scientific output in this field has grown each year to a cubic power law $\sim t^3$. Given that this output is growing at a rate that is slightly above the overall graphene

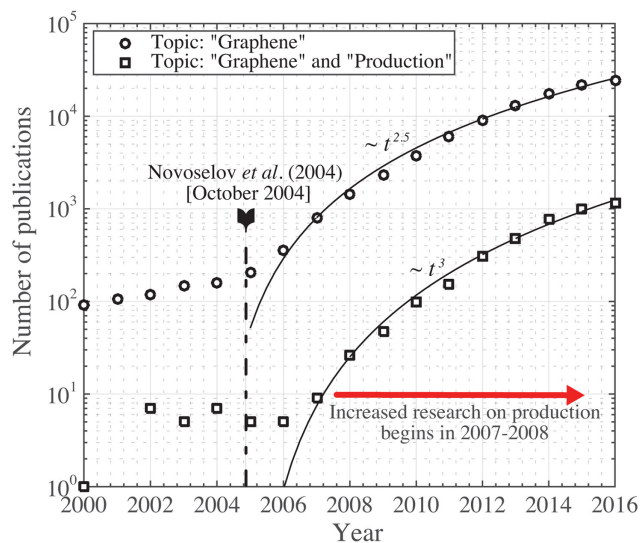


Figure 2. The trends in number of publications for Graphene and Graphene Production topics between 2000 and 2016.

Data taken from Figure 1a). [Color figure can be viewed at wileyonlinelibrary.com]

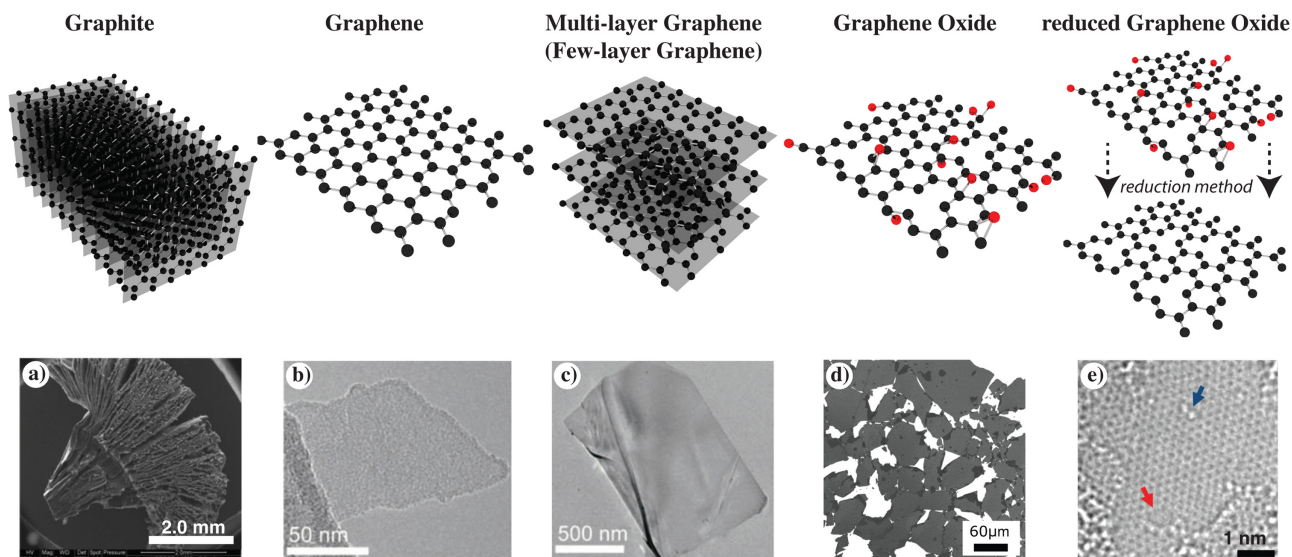


Figure 3. Overview of graphite and the various types of graphene-based materials with corresponding representative microscopy images.

a) Scanning Electron Microscopy (SEM) image of expanded highly orientated pyrolytic graphite from Cooper et al. (b–c) TEM images of mono- and multilayer graphene from Qian et al. d) SEM image of GO from Voiry et al. e) High resolution TEM (HRTEM) image of single layer reduced graphene oxide with indications of holes (red arrow) and oxygen functional groups (blue arrow) from Voiry et al. Reproduced with permission from Ref. 15–17. Copyright 2014 Elsevier B.V., Copyright 2009 Springer, Copyright 2016 American Association for the Advancement of Science. [Color figure can be viewed at wileyonlinelibrary.com]

research output, the outlook for continued development and refinement of production approaches is encouraging.

A breakdown of the generation of intellectual property (IP) through patents goes some way toward understanding the importance of graphene production to the industrial sector. Figure 1b presents a noncomprehensive gathering of the number of patent applications, providing a sample view on the landscape. In contrast to the scientific output, production aspects are a much higher fraction at almost 15% of the general graphene IP area in 2015. The difference between scientific output and generation of IP is not surprising, given that mass-production is a topic directly linked with industrial practices. Economic factors may also have an influence. A recent forecast for 2017 predicts that material supply will account for \$75M of a >\$250M graphene/2-D material global market, which includes end-use applications.¹⁴ As the graphene technology space develops, it is expected that a redistribution in IP will occur from production/synthesis towards end-use applications of graphene and graphene-related materials.¹⁰

Graphene-Based Materials

The type of material produced by bottom-up and top-down approaches varies depending on the production method. In Figure 3, schematic representations of these different types of materials are provided together with corresponding electron microscopy images of their respective morphologies. Bottom-up approaches typically produce monolayer graphene and multilayer graphene (MLG, $2 < N_1 < 10$), with highly controlled layer numbers (N_1). Top-down approaches utilize graphite ($N_1 > 10$) and exfoliate this precursor into monolayer and multilayer graphene with a broad distribution in layer number. Top-down approaches which implement chemical and electrochemical exfoliation mechanisms produce graphene oxide (GO), a form of graphene that includes a high oxygen content resultant from the exfoliation process. Further postprocessing of GO, to reduce the oxidation state and

increase electrical and thermal conductivities, produces reduced graphene oxide (rGO). In these forms, GO and rGO can be useful for certain applications such as energy storage and composite materials.¹⁰ There is an intrinsic relationship between the different production methods and the type of graphene-based material that is produced, as illustrated in Figure 4. Although the focus of this review is on graphene production methods, the reduction approaches discussed below have been included for completeness.

Existing Production Routes

The “bottom-up” and “top-down” approaches are illustrated in Figure 4. Graphene of high quality can be produced through bottom-up approaches, which is suited for use in electronic devices. These techniques, however, have generally suffered from very low production rates. In contrast, top-down processes, which most commonly use liquid-phase exfoliation, produce graphene at much higher production rates but often result in noticeably lower quality. The restacking of the sheets produced is commonly restricted using suitable solvents or aqueous surfactant solutions (see section on Graphene Dispersion). The resulting graphene has been typically reserved for composites, conductive inks, coatings, and flexible electronics applications.

Providing direct comparisons between the two production routes is challenging. Each route holds particular advantages and disadvantages in terms of use of feed materials and energy, preprocessing and postprocessing requirements, graphene product specifications, technology and process complexity, environmental sustainability, and many other important aspects. Similarly, the properties required for the end-use application may determine the choice of production route. There are many applications where graphene of the highest quality and aspect ratio (lateral size to thickness ratio) is unnecessary. Considering the breadth of applications mentioned previously, it is probable that top-down approaches

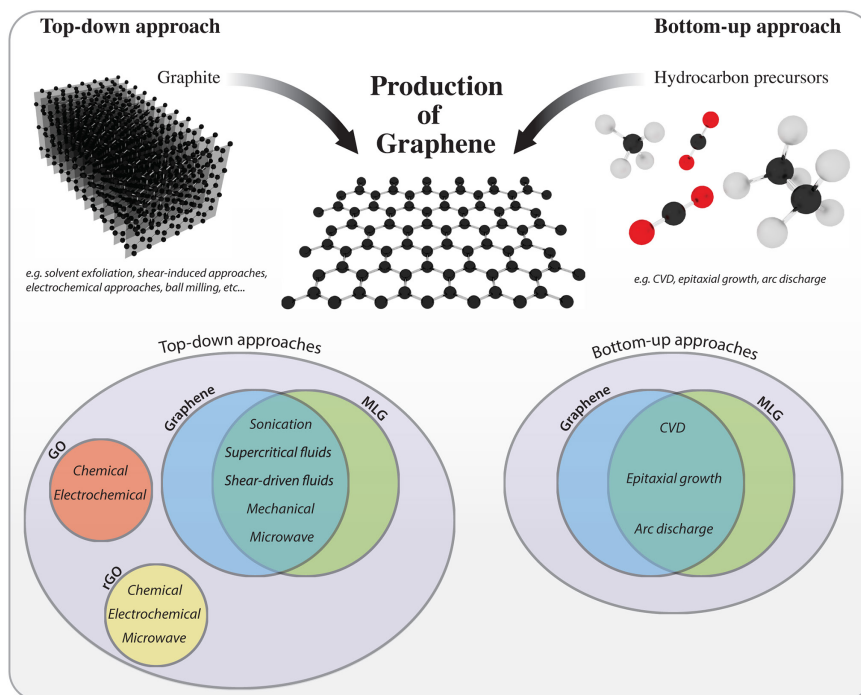


Figure 4. Top-down and bottom-up routes to graphene production.

The intrinsic relationship between production (also reduction) approaches and graphene-based material type is also shown. [Color figure can be viewed at wileyonlinelibrary.com]

will provide the primary source of mass-produced graphene in the near future. The demand for graphene produced via top-down methods will, therefore, be influenced by the broad interest in graphene-based technological developments from various industries. If there is going to be a stable and growing demand for a lower quality graphene produced via top-down methods, it should encourage graphene producers to develop novel top-down processes, with the aim of improving process intensification and material quality.

Furthermore, top-down approaches principally offer more financially feasible production (continuous or batch) of graphene at a large-scale relative to bottom-up methods. Many of the top-down methods developed can be easily integrated with other widely used processes to either modify graphene for distinct applications, or to use it directly in processes such as chemical functionalisation, surface modification, and film deposition.¹⁸ Despite these process integration benefits, the most widely researched methods such as liquid-phase and electrochemical exfoliation, require substantial improvements in process efficiency, sustainability, and monolayer production yield, for feasible scale-up.

When choosing to further advance a particular graphene production route, it is preferential to evaluate key characteristics such as the production yield and throughput. An example of such evaluation is presented in Figure 5, which summarizes the data reported in a selection of studies on graphene production via top-down and bottom-up methods.^{16,19–31} This type of dataset highlights key production outputs, and provides a clear comparison between the methods based on their important features. For instance, the arc discharge method can potentially produce graphene at relatively high rates, but it suffers from having low production yield. If one can consider advancing or optimising process conditions, then this could probably lead to an improvement of graphene production yield. Similarly, methods that produce graphene at a high yield

but at low rate, for example, microwave and ball milling, can be evaluated to understand the relationship between both variables and increase throughput. Although those methods that suffer from both low production rate and yield present risks from a technological viability perspective, these may also give opportunities to explore the production of targeted products, for example, perforated graphene via pressure driven methods,

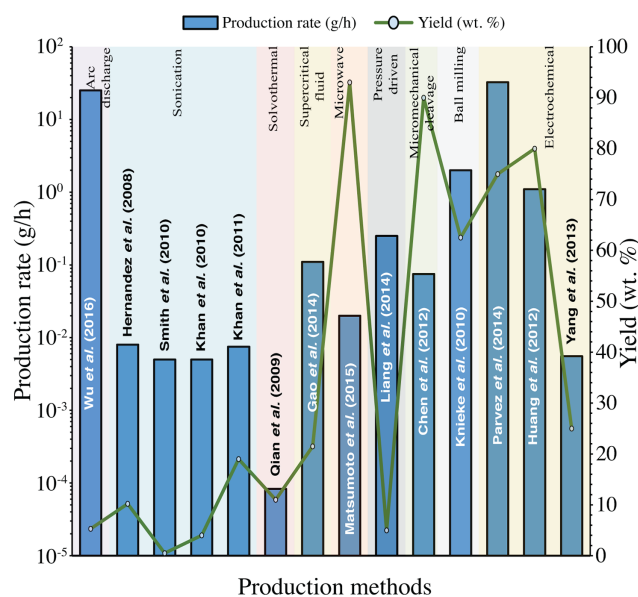


Figure 5. Summary of graphene production methods.^{16,19–31}

Criteria for data selection: a method results in at least 50% production of graphene with ≤ 5 layers, and has reported values for yield (wt %) and production rate (g/h). [Color figure can be viewed at wileyonlinelibrary.com]

or use of sonication for postproduction and storage. As noted previously, such analysis must encompass other key measures such as graphene defects, selectivity, target properties, input, and operation costs.

Liquid-Phase Exfoliation

The top-down route to production of graphene encompasses a wide range of methods that use a graphite precursor as a starting material. Liquid-phase exfoliation is the most common area of research in this field, where the graphite precursor is exfoliated directly within a liquid. This section explores the key factors that impact the scale-up of production including the precursor, the dispersion medium, the exfoliation mechanisms, and the current techniques. Three main approaches that have emerged over the past decade are discussed. These have been grouped into Mechanical, Chemical, and Electrochemical exfoliation.

Precursor

Graphite is the main raw material (apart from other chemical precursors and dispersants, see section Graphene Dispersion) used in top-down methods of graphene production. Graphite itself is produced either naturally or synthetically, and consists of polycrystalline particles or granules which are composed of multiple single crystals. In natural graphite, these crystals are orientated in a preferred direction, whereas the orientation in synthetic graphite is more random. Graphite consists of multiple 2-D graphene layers, held in parallel to each other by van der Waals forces, forming a 3-D crystal. These layers are organized to create a stable structure with a particular stacking sequence. The most common sequence is hexagonal (ABAB...), whereas rhombohedral (ABCABC...) is rather rare.³²

Both natural and synthetic graphite materials have many different varieties, for example, “amorphous” graphite, flake graphite, vein graphite, high purity flake graphite, expanded graphite, highly oriented pyrolytic graphite (HOPG), Kish graphite, and so forth.³³ Different types of graphite of

particular grain sizes and purities have been used by other researchers, and this information is displayed in Table 1. All of these types of graphite variations have differences, either small or large, in their physical and chemical properties. These differences include particle-size distribution, morphology, and purity (see Figure 6 for a more detailed breakdown). The properties play a critical role, not just in the control of the exfoliation process and the quality of the graphene output, but the applicability of the graphene in downstream industrial processes. It is necessary to have a thorough understanding of the raw material characteristics, as the fundamental exfoliation mechanisms are occurring at the nanoscale.

Five different types of graphite materials (HOPG, natural graphite flake, Kish graphite, flake graphite powder, artificial graphite) were used by Wu et al. to produce graphene using a chemical exfoliation method. These materials have different lateral sizes, thicknesses, morphologies, and purities. It was found that the selectivity of the number of graphene layers is strongly correlated with the lateral size and crystallinity of the precursor graphite. As the lateral size and crystallinity of precursor graphite decreases, the thickness of the produced graphene also decreases.⁵²

Similar observations were made by Knirsch et al. in a study on the chemical reactivity of three different graphite sources. This included graphite flakes, graphite powder, and spherical graphite. All three precursors have different morphologies, density of defects, and bulk densities. Their analysis showed that powder graphite yields the highest degree of functionalization—(I_D/I_G) average = 3.1. They also confirmed that larger graphite flakes undergo only partial exfoliation, due to higher intrinsic lattice energy.⁵³ The significance of natural graphite particle size was also carefully studied by Arao et al. It was shown that the production of few layer graphene is much more efficient when small graphite particles are used relative to larger graphite particles. Their findings also add to the fact that the energy required for exfoliation increases with increasing graphite particle size.⁵⁴

Table 1. Summary of Different Graphite Precursor Used in the Literature to Produce Graphene

Type	Grade	Purity	Size	Ref.
Graphite flakes, natural	n/a	n/a	+100 mesh ($\geq 75\%$ min)	34
Graphite flakes, natural	n/a	n/a	+100 mesh ($\geq 75\%$ min)	35
Graphite flakes, natural	n/a	96%	>32 mesh	35
Graphite flakes, natural	n/a	n/a	+100 mesh ($\geq 75\%$ min)	21
Graphite flakes, natural	n/a	99.80%	325 mesh	36
Surface enhanced flake graphite, natural	Grade 3725	n/a	<3 μm	37
Graphite flakes, natural	HP230U	99.60%	20 μm	38
Expandable graphite, natural	n/a	99%	average diameter of 300 μm	16
Expandable graphite, natural	Grade 3772	99.35%	20 $\times$ 50 mesh	37
Expanded graphite, natural	n/a	99.50%	10–30 μm	39
Graphite powder, natural	n/a	n/a	0.5 mm mesh	40
Graphite powder, natural	n/a	n/a	<0.1 mm	41
Flake graphite powder, natural	n/a	> 99.9%	25 μm	42
Powdered graphite, natural	n/a	$\geq 99.99\%$	$\sim 70 \mu\text{m}$	43
Graphite powder, natural	n/a	$\geq 99.99\%$	100 mesh	44
Crystal graphite powder, natural	n/a	$\geq 98.0\%$	≤ 300 mesh	45
High purity graphite rods, natural	n/a	$\geq 99.99\%$	diameter 6 mm	46
Primary artificial, synthetic	Grade TC307	$\geq 99.9\%$	<1 μm	37
Graphite powder, synthetic	n/a	$\geq 99.99\%$	1–10 mm	47
Graphite powder, synthetic	n/a	n/a	<20 mm	48
Graphite powder, synthetic	SP-1	$\geq 99.99\%$	100–200 mm	47
HOPG	SPI-2 grade	≤ 10 ppm ash	10 $\times$ 10 $\times$ 1 mm	15
HOPG	SP-1	n/a	10 mm $\times$ 10 mm $\times$ 0.2 mm	49
HOPG	n/a	n/a	1 $\times$ 1 cm^2	50
HOPG	n/a	n/a	n/a	51

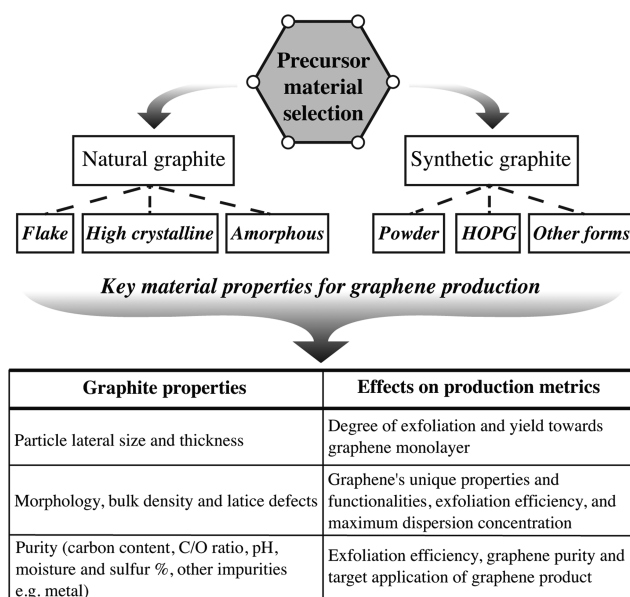


Figure 6. Summary of the graphite precursor considerations and effects on graphene production.

Kozhemyakina et al. used 22 different graphite grades to study their dispersibility in various solvents. This study allowed them to clearly identify that the smaller graphite particles (relative to larger ones) can be more readily dispersed in a solvent and form a more stable dispersion. This can be even further enhanced if graphite of small bulk density and pH close to neutral in water is used.⁵⁵

Considering other graphite properties, the presence of impurities such as iron in flake and Kish graphite will result in paramagnetic properties, which can be either beneficial or undesirable as a precursor material for particular applications. For instance, graphene with iron could be used in the manufacture of composites and thin film conductors, but it would be undesirable to use it in spintronic or magnetic applications.⁵⁶

Graphite can also be pretreated to encourage exfoliation, and potentially increase the production yield of high quality graphene with high aspect ratio. An intercalation process could be used to produce expanded flake graphite. This pretreated precursor could then be used in an exfoliation process to produce graphene with beneficial properties, such as high electrical conductivity.³⁷ This pretreatment, however, should be thoroughly controlled and assessed to mitigate any unwanted changes in important graphene properties. This topic requires a comprehensive research effort, to understand the defects introduced, how these arise and how these could be potentially healed.⁵⁷ If graphene of very high quality and purity is required, such as in the electronics industry, HOPG is one of the desirable starting materials. The limited supply of HOPG, however, makes it currently financially unviable for large-scale production.

It is crucial, therefore, that graphene producers at least consider the type and grade of graphite to be used, including grain size, density and purity. Control over the precursor characteristics translates into a finer control over the exfoliated product. This can also benefit downstream industrial use of graphene produced using top-down methods. Employing collaborative research efforts, between graphite producers in industry and academic institutions investigating top-down production methods, could be a key driver in the advancement of optimal, economically viable precursors.

Graphene Dispersion

In addition to the precursor material, another key component of the exfoliation process is the selection of an appropriate liquid medium. Graphene concentration benefits when suitable liquids are used to separate nanosheets from the graphite precursor, and inhibit the thermodynamically favorable aggregated state for long periods of time. Solvents and aqueous surfactant solutions are the predominant routes for the metastable dispersion of pristine graphene in nonoxidative liquid phase exfoliation techniques. In addition, an approach known as “Reductive Dissolution” has been shown to produce thermodynamically stable liquid formulations of nanocarbons. This chemical method spontaneously deconstructs a graphite intercalated compound into its constitutive graphene layers. A brief summary of this dispersion approach has also been included. In all cases, there has been an increased effort over the past decade to characterize the dispersive performance of liquids, and obtain fundamental information on molecular-scale interactions.

Solvents

Solvent-based exfoliation of graphene has become the most common route in liquid-phase exfoliation production methods. This approach to exfoliation of graphene has its origins from earlier work on the exfoliation and dispersion of carbon nanotubes.^{58–62} Challenges in aggregation and bundling of single-walled carbon nanotube (SWCNT) powders were addressed through dispersion in suitable solvents. These solvents balance the attractive potentials with repulsive potentials, preventing aggregation and stabilizing the mixture, once the bundles of nanotubes are exfoliated. This aggregation is a fundamental result of van der Waals interactions between carbon atoms in adjacent nanotubes.⁵⁹

The physics of this mutual attraction is similar for graphene nanosheets. Additionally, the similarity between recently proposed solvent parameters⁶³ and surface energies⁶⁴ suggested that solvents used in SWCNT studies could also be used to exfoliate graphite into graphene. This is the reason, therefore that many of the “good” solvents originally used in SWCNT dispersion are also common in the more recent research on liquid-phase exfoliation of graphene. Toward the end of the last decade, a comprehensive experimental campaign was undertaken to analyse the effectiveness of numerous solvents to exfoliate and disperse graphene.^{21,63} Hernandez et al.⁶³ presented graphene concentration data for 40 different solvents. Cyclopentanone (CPO) provided the highest concentration, and N-Methyl-Pyrrolidone (NMP) was also a high performing solvent. The latter has become one of the most popular solvents in liquid-phase exfoliation studies to date. Water was a poor solvent for graphene exfoliation and dispersion, positioned 35 out of 40, and providing concentrations 1/8 of that achieved with CPO. This highlights one of the primary challenges in scaling-up solvent-based liquid-phase exfoliation. Many of the good solvents, such as NMP, have a high boiling point and are toxic.

In an effort to correlate the experimental findings, and provide a physical interpretation behind the solvent ordering, the authors considered surface energy as a parameter²¹ along with Hildebrand and Hansen solubility parameters.⁶³ The overarching idea, linked to similar work on SWCNTs,⁶⁵ is that suitable solvents can be selected through minimization of the enthalpy of mixing. The initial work proposed that matching solvent and graphite surface energy would minimize the energetic cost

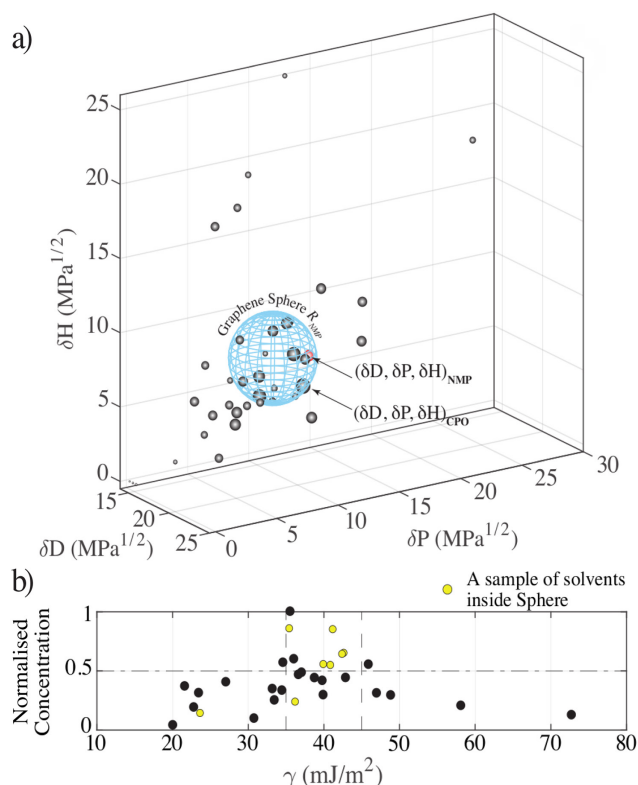


Figure 7. a) Hansen space representation and b) surface tension plot of solvents used for exfoliation and dispersion of graphene. Experimental data from.⁶³

[Color figure can be viewed at wileyonlinelibrary.com]

of exfoliation. The recommendation was the use of solvents with surface tension close to 40 mJ/m². In a follow-on study, Hernandez et al.⁶³ suggested that although surface tension is a useful parameter to assess the overall interaction, the intermolecular interactions could be divided into at least three types based on the Hildebrand and Hansen solubility parameters. The cost of mixing solute-solvent was related to the Flory-Huggins interaction parameter, which scales as $\chi \sim R^2$. Here, R represents the Hansen solubility sphere between solute (1) and solvent (2), and $R^2 = 4(\delta D_1 - \delta D_2)^2 + (\delta P_1 - \delta P_2)^2 + (\delta H_1 - \delta H_2)^2$. The dispersive, polar, and hydrogen-bonding components are defined by δD , δP , and δH respectively.⁶⁶ The cost of mixing is, therefore, reduced when small values of R are achieved. This is the commonly known “like dissolves like” concept.

Hansen parameters are regularly used in molecular solubility analyses, and their applicability to treat nanomaterials (with well-defined surfaces) as a classical solute remains questionable. To illustrate this, Figure 7a presents the parameters for the solvents from the previous graphene exfoliation studies in three-dimensional Hansen space. The radius of each data point is scaled with respect to graphene concentration after centrifugation. The Graphene Sphere R_{NMP} represents the relative energy distance for graphene and NMP as a solvent. The solubility parameters for graphene were estimated as $\langle \delta D \rangle \approx 18$, $\langle \delta P \rangle \approx 9.3$, $\langle \delta H \rangle \approx 7.7$ (units: MPa^{1/2}).⁶³ This example was chosen given the widespread use of NMP in the liquid-phase exfoliation of graphene.

Taking a rigorous interpretation of the Hansen solubility sphere, if NMP is a good solvent than any other solvent within

this sphere should also be equally good (or better) for exfoliation and dispersion of graphene. In this case, only six solvents of the top 10 are within this sphere (Figure 7b), with the best solvent CPO outside. There are also poorer solvents, such as Pyridine, within this bounded space. Similarly, there is a scatter of good and poor solvents that have a surface tension close to 40 mJ/m². Figure 7b implies that if a particular solvent is to be classified as good, then it will most likely have a surface tension within the 35–45 mJ/m² range. Put another way, it is not guaranteed to be a good solvent if it has the recommended surface tension of 40 mJ/m². The same probabilistic interpretation appears to also apply in the application of Hansen solubility parameters to graphene-solvent dispersions. This suggests that, although these empirical reduction methods hold merit in guiding the search for suitable solvents to exfoliate and disperse, fundamental insights at the material-solvent interface are required.

Molecular-scale simulations of monolayer graphene sheets in solvents provides the opportunity to study this material-solvent interface in detail. Shih et al.⁶⁷ produced a thermodynamic description of the stability of graphene in five polar solvents (NMP, DMSO, DMF, GBL, and water) using Molecular Dynamics (MD). Two parallel monolayer sheets were simulated in a domain comprising the solvent of interest. The distance between both sheets, d , was varied to understand the interaction. This interaction, when using NMP as a solvent, is shown in Figure 8. The solvent has two beneficial characteristics which both aid exfoliation concentration and reduce aggregation. First, the van der Waals energy well is shallower than that for two graphene sheets in vacuum. Second, a high energy barrier exists, which must be overcome if graphene sheets are to aggregate. Comparing the potential of mean force (PMF) with number of NMP molecules as distance is

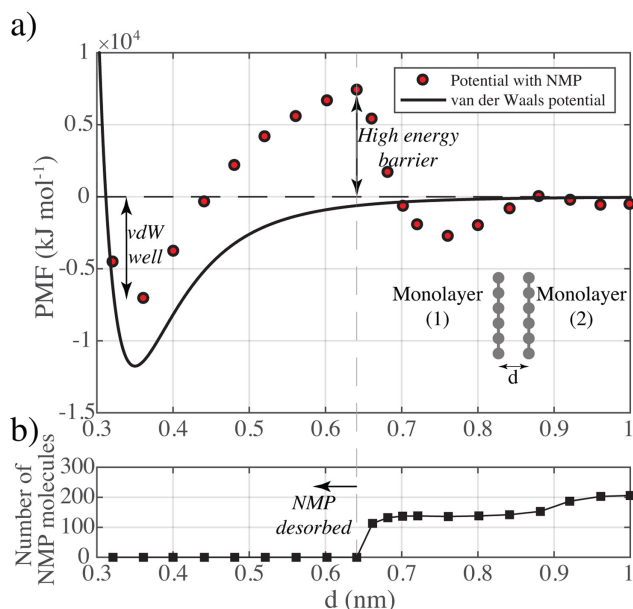


Figure 8. a) Comparison between the potential of mean force for NMP, predicted by MD simulations of two graphene monolayers,⁶⁷ and the classical Lennard-Jones (van der Waals) interaction potential in vacuum. b) The number of NMP molecules confined between the two graphene monolayers.⁶⁷

[Color figure can be viewed at wileyonlinelibrary.com]

decreased, the rapid change in peak energy at $d = 0.66$ nm corresponds to the desorption of a single layer of NMP molecules that are positioned between the graphene sheets. This occurs when the total potential energy of the confined solvent molecules exceeds the bulk solvent–solvent potential energy. The van der Waals interaction between the graphene sheets, therefore, dominates in the absence of NMP molecules at $d < 0.66$ nm.

As a way to illustrate why certain solvents are good/poor, based on the findings of Shih et al.,^{6,7} Figure 9a presents the PMF per unit area for NMP (good), DMF (good), and water (poor) examples. Both NMP and DMF have similar magnitudes for the van der Waals energy well and the high energy

barrier, with NMP performing slightly better. Graphene can remain dispersed, even though the ultimate thermodynamic equilibrium state is in the aggregated form (i.e., graphite). The 2-D sheets can be held in a kinetic or metastable state for a long period of time if this energy barrier is sufficiently large.⁶⁸ The larger the energy barrier, the slower the kinetics, which explains why NMP has been shown experimentally to be one of the most suitable solvents for long term dispersion. In contrast, the depth of the energy well for water is almost double that of the good solvents, and the energy barrier is extremely low.

Shih et al.⁶⁷ also produced a fit of the numerical data, to capture the effect of the confined single layer of solvent molecules. The fit is based on the classical Lennard-Jones potential, with additional terms to model the energy barrier and secondary energy well contributions. This is shown in Figure 9a, and provides a useful insight for the molecular design of solvents that can enhance exfoliation and dispersion. The PMF results were combined with theory of slow colloid coagulation, to compare time-dependent aggregation observations with experiment. In NMP, the fraction of monolayer graphene can reduce from 27% to just 9% after 10 days as sheets aggregate. This highlights the considerable challenge in enabling long-term storage, as NMP is currently one of the most efficient solvents for liquid-phase exfoliation. Overall, the predictions provided a fundamental insight into the molecular stabilization mechanisms and a practical tool for the determination of time-dependent aggregation in solvents.

Surfactants

Surfactants can be used to lower the surface tension of water, making this environmentally benign solvent suitable for the dispersion of graphene. Surfactant-based exfoliation and dispersion addresses some of the disadvantages associated with the use of solvents. Aqueous dispersions using surfactants have a low boiling point, compared to the widely used solvents, and also have environmental benefits. One of the limitations, however, is the adverse effect residual surfactant can have on the electronic properties of graphene. This can result in a need to remove the surfactant in a postprocessing step, such as immersion in nitric acid for a period of time and rinsing with deionized water.⁷⁰ A positive impact is that the electronic structures can be fully recovered. The negative is that a further processing step is required with additional chemicals, leading to increased cost in scale-up. Unsurprisingly, the choice of surfactants in liquid-phase exfoliation of graphene also originated from the earlier work on carbon nanotubes.^{61,71,72} The Vigolo et al. study⁷¹ was one of the earlier works that assessed the influence of the surfactant sodium dodecyl sulphate on the dispersion performance of carbon nanotubes in water solution. Phase diagrams were used to show the influence of both surfactant and carbon nanotube concentrations on the level of homogeneity in the dispersion. Lotya et al.⁴⁰ used sodium dodecylbenzene sulfonate (SDBS) to demonstrate exfoliation and dispersion of graphene using sonication. The relatively stable dispersions were then used to produce films with vacuum filtration and spray coating.

Surfactants can promote graphene dispersion through the formation of an electric double layer, or alternatively, a steric stabilization effect.⁶⁸ An electric double layer is repulsive, preventing the attractive van der Waals potential from aggregating the individual monolayer or few-layer graphene sheets. The repulsive electrostatic potential for two monolayer sheets

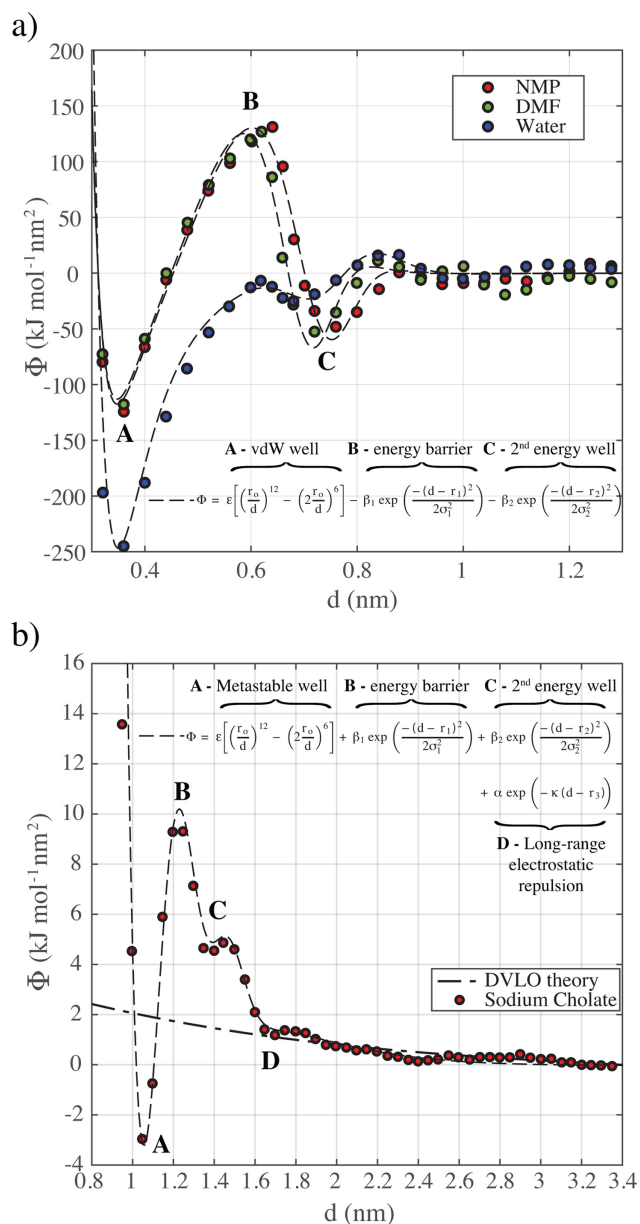


Figure 9. Interaction potentials, represented by the potential of mean force per unit area, between two graphene sheets in a) NMP, DMF, and Water, and b) in an aqueous surfactant solution using Sodium Cholate (SC).

Data points are MD simulation data from Ref. 67,69. [Color figure can be viewed at wileyonlinelibrary.com]

separated by distance, d , is presented in Figure 9b using DLVO theory.^{68,69}

A steric repulsion can be formed when the hydrophobic tail of a surfactant molecule adsorbs to a graphene sheet. The hydrophilic component extends into the water, and interacts with other hydrophilic parts when surfactant coated graphene sheets come into close proximity.²⁹ This compressive interaction has an unfavourable entropy, resulting in a steric repulsion potential.⁶⁸

The quantity of monolayers produced with surfactants was found to be lower than that achieved using solvent based exfoliation.⁷³ Lotya et al.⁷⁴ produced a maximum of 0.3 mg/mL of graphene for a prolonged 430 h of sonication time. The optimal concentration of sodium cholate (SC) in water was found to be 0.1 mg/mL. With the same surfactant concentration, Smith et al.²⁹ tested 12 ionic and nonionic surfactants for direct comparisons. The comparison between both types of surfactants also provided an insight into both the electric double layer and steric stabilization mechanisms. The length and thickness of the exfoliated sheets was similar for all, however, concentration ranged between 0.011 and 0.026 mg/mL. Part of this variation may have also been related to the application of nonoptimal surfactant concentrations. In 10 of the 12 cases studied, the concentration was well below the critical micelle concentration.⁶⁸ One advantage of this, however, is that it results in less surfactant removal during the postprocessing steps. Overall, there appeared to be minimal differences between the maximum concentrations achieved using ionic and nonionic surfactants. Guardia et al.⁴¹ however, measured substantial improvements using nonionic over ionic surfactants. Concentrations of graphene were also produced at much higher levels up to approximately 1 mg/mL. Given the impact of surfactants on production output and material electronic properties, there has been a desire to optimize surfactant concentrations.⁷⁵ Sun et al.⁷⁶ maximized graphene concentrations up to 7.1 mg/mL using sodium taurodeoxycholate and tip sonication (90% graphene < 5 layers).

Even higher graphene concentrations, up to 50 mg/mL, have been obtained using a flavin mononucleotide (a derivative of vitamin B₂) stabilizer in water.⁷⁷ Biomolecule-assisted exfoliation is another relevant field that has recently emerged.⁷⁸ Dispersion of graphene has been achieved with biomolecules including proteins and peptides, RNA, DNA, plant extracts, and bile salts. The principle dispersion mechanisms are similar as that discussed here for surfactants. Further details on this area of research and end-use applications have been reviewed recently.⁷⁸

Simulations of monolayer sheets in aqueous surfactant solutions have also led to an improved understanding of surfactant dispersed graphene at the molecular scale.⁶⁹ Figure 9b describes the interaction potentials with sheet separation distance for two graphene monolayer sheets. These potential mean force predictions are the equivalent to that shown in Figure 9a for polar solvents. A deviation between DLVO theory and the molecular dynamics predictions occurs in the near-field region, where $d = 1.6$ nm. Beyond $d = 1.6$ nm, there is reasonable agreement, as the long-range electrostatic repulsion is the interaction potential which dominates. This electrostatic contribution is captured by DLVO theory.⁶⁸

In the near-field, an energy barrier is produced by steric repulsion, as the surface adsorbed cholate ions come into contact with each other. A metastable well, at $d = 1.05$ nm, features a single layer of cholate ions and a sodium-ion wall

confined between both sheets.⁷⁹ Most notably, this study highlights that the electrostatic interaction is not the dominant repulsive force as assumed previously for ionic surfactants.²⁹ This comes primarily from the steric repulsion in the last layer of sodium cholate (SC) confined between the graphene sheets. The inset model, fitting the data in Figure 9b, is similar to that proposed in the original study on solvents.⁶⁷ This potential mean force model was ultimately used to predict the time-dependent concentration of graphene using colloid aggregation theory. This forms a useful engineering tool for estimating monolayer lifetimes in aqueous surfactant dispersions. Graphene concentrations in SC have been found to reduce by almost 70% in 10 days.⁶⁹ Predicted monolayer number fractions become depleted, as more few-layer sheets ($N > 3$) are formed through aggregation. As with solvent dispersions, one of the main process engineering challenges will be to maintain an as-exfoliated layer distribution throughout long-term storage.

Reductive dissolution

Once the liquid phase exfoliation mechanism is stopped, both traditional solvent and surfactant-based dispersions of graphene ultimately return to an aggregated state over time due the conditions being thermodynamically unstable. An alternative chemical processing approach is reductive dissolution, which wholly deconstructs graphite particles into their constitutive graphene flakes and maintains dispersion in a liquid.⁷⁹ This feature yields a stable, spontaneously soluble graphene product, which the previous dispersion mechanisms have yet to demonstrate.

The dissolution process uses a graphite intercalation compound (GIC), such as potassium graphite KC₈, as the precursor material. Electrostatic interactions occur between graphene layers, and these two-dimensional negatively charged flakes are dissolved spontaneously in a polar solvent such as NMP. This occurs as the favourable entropic term ($T\Delta s$) in the free energy of mixing relationship ($\Delta G = \Delta h - T\Delta s$) dominates the unfavorable enthalpic contribution (Δh), similar to that described for the dissolution of nanotubes.^{79,80} Due to the charged nature of the material, leaders in this field have labelled the product as graphenide⁸¹ to distinguish it from neutral graphene. This follows the terminology introduced for dissolved carbon nanotubes (nanotubides⁸⁰) and fullerenes (fullerides⁸²).

The spontaneous dissolution process has a number of advantages as a production route. Both quality and quantity are high relative to the methods of mechanical exfoliation in liquids. Lateral flake sizes are large (50 μm shown in Ref. 81), as the dissolution process inherently avoids the fragmentation mechanisms that are common in other liquid phase exfoliation methods such as sonication. The approach has been shown to work with other low boiling point solvents,⁸³ thus addressing the process and environmental challenges associated with scaling up production with the commonly used solvents discussed in a previous section on solvent dispersion. A disadvantage, however, is that the solutions must be processed in an inert atmosphere. The effects of the intercalation process, have been shown to be relatively minor, and the work function can be similar in magnitude to that of undoped graphene.

Recently, negatively charged graphene has been dispersed in water without surfactant using the product derived from this reductive dissolution method.⁸⁴ In this breakthrough, graphenide and tetrahydrofuran were mixed with degassed water.

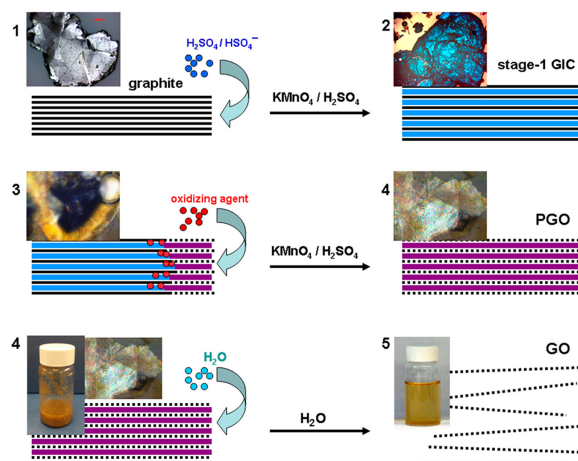


Figure 10. Schematics of the conversion of bulk graphite into graphene oxide with corresponding micrographic images or sample appearances at each phase.

The three steps indicate the formation of the two intermediate products (graphite intercalated compound, GIC, and pristine graphite oxide [PGO]) and the final GO product. The solid black lines represent graphene layers; dotted black lines represent single layers of GO; wide blue lines represent $\text{H}_2\text{SO}_4/\text{HSO}_4^-$ intercalant; wide purple lines represent a layer of the mixture of $\text{H}_2\text{SO}_4/\text{HSO}_4^-$ intercalant with the reduced form of oxidizing agent. Reproduced with permission from Ref. 87. Copyright 2014 American Chemical Society. [Color figure can be viewed at wileyonlinelibrary.com]

The organic solvent was evaporated off to leave graphenide dispersed in water at a concentration of 0.16 g/L. For further information on the reductive dissolution of graphene, and other nanocarbon materials, a detailed description is provided elsewhere.^{79,81}

Exfoliation Mechanisms

Exfoliation of graphite in a liquid can be achieved through various routes. The fundamental mechanisms used in most of the current top-down exfoliation techniques, can be broadly divided into three main areas. These principal mechanisms are mechanical, chemical and electrochemical. Mechanical exfoliation methods are typically nonoxidizing, meaning the graphene is produced without functionalization of oxygen groups that can adversely impact electronic properties. Electrochemical exfoliation methods may, in certain circumstances, have an indirect oxidizing effect. The level of oxidation (i.e., C/O ratio), however, is limited compared to strongly oxidizing chemical approaches which often require postprocessing reduction steps to improve material properties.

Mechanical exfoliation

Mechanical exfoliation relies on the use of force to break the van der Waals attraction between the planes in graphite. There are essentially three ways to overcome this attraction⁸⁵:

1. Peel two graphite layers apart by applying a normal force to overcome the van der Waals attraction.
2. Application of a lateral, or shear force. Adjacent layers within graphite can slide in the lateral direction, when a suitable force is applied to overcome the van der Waals attraction.
3. Use exfoliation with fragmentation to break-down large graphite layers into smaller sizes. If the desired product does not require graphene of large area, then the method promotes easier exfoliation due to lower van der Waals interaction forces that are split across smaller sized graphite flakes.

Chemical exfoliation

Chemical exfoliation techniques utilize chemical approaches to turn the starting graphite into a graphite derivative. This can then be readily delaminated to give single or few layer sheets of chemically derivatized graphene. This derivative can receive subsequent treatment to convert it back to graphene. The most common of these techniques is the graphite oxide route, which capitalizes on strongly oxidising graphite to produce graphite oxide. There are many approaches that have been developed for the preparation of graphite oxide. The Brodie method was developed using fuming HNO_3 and KClO_3 as intercalants and oxidants.⁸⁶ Hummers and Offeman proposed a method in which H_2SO_4 , NaNO_3 , and KMnO_4 were used. This method eliminated some of the hazardous steps that were present. However, the key issues still limiting the adoptability of this method includes the formation of toxic gases such as $\text{NO}_2/\text{N}_2\text{O}_4$, difficulty in extracting out Na^+ and NO_3^- , and low conversion of graphite to graphite oxide.⁸⁶

The mechanism for the formation of graphite oxide, and subsequent treatment in solution to form graphene oxide, can be seen in Figure 10. Graphite is first oxidized using strong oxidants and with further hydrolysis, graphite oxide can be separated to give graphene oxide.

The chemical exfoliation process affects the final graphene oxide product by generating irreversible defects on graphene sheets during the oxidation process. This makes it insulating due to the disruption of graphitic networks. Graphene oxide then needs to be reduced to remove the oxygen functional groups, making it useful for electronic applications.⁸⁸ The final quality of the reduced graphene oxide, therefore, is dependent on the experimental parameters that are applied in the reducing step. The number of defective sites is nonetheless often—although not always—higher in rGO compared to other forms of graphene.⁸⁹ There are various reduction methods that are available in the literature for making rGO. The main ones include chemical reduction and thermal reduction. Other techniques such as solvothermal, photocatalytic, biomolecules, natural agents, and plasma assisted reduction have also been explored. Table 2 summarizes some of these

Table 2. List of the Reduction Methods of Graphene Oxide Along with Key Features of the Resulting Reduced Graphene Oxide. Reproduced with permission from Ref. 91. Copyright 2015 Elsevier B.V

GO Reduction Methods	Reduction Processes	C/O ratio	Electronic Properties	
			Mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	Conductivity (S/m)
Chemical reduction	Hydrazine	10.3 ^b	—	2420
	Phenylhydrazine	9.5 ^b	—	4700
	$\text{NH}_3\text{-BH}_3$	9.8 ^a	—	20300
	55% HI acid	12.0 ^a	—	29800
	HI-AcOH	11.5 ^b	—	30400
	Ethylenediamine	7.8 ^a	—	220
	Na-NH ₃	16.6 ^a	123 (hole)	—
	Nascent hydrogen (Al/HCl)	21.1 ^b	—	12530
	HI + trifluoroacetic acid	10.5 ^a	250 (hole) 200 (electron)	—
	Zn/HCl	33.5 ^a	—	15000
Thermal reduction	Thiourea dioxide/NaOH	14.5 ^a	—	—
	700°C, H ₂ , 30 min	28.6 ^a	—	8100
	1000°C, H ₂ , 1 h	—	5.4 (hole) 1.1 (electron)	76000
	1000°C, C ₂ H ₂ , 30 min	50.2 ^a	—	143000
Plasma-assisted reduction	CH ₄ plasma, 700°C, 20 s	9.2 ^a	—	34500
	Ar/H ₂ plasma, 150°C, 30 min	7.0 ^a	—	—
Photocatalytic reduction	Polyoxometalate cluster ($\text{H}_3\text{PW}_{12}\text{O}_{40}$)	—	0.03 (hole) 0.01 (electron)	120
Solvothermal reduction	DMF + 0.6 M H ₂ SO ₄ , 1 h	8.4 ^a	—	1223
	H ₂ O:NMP (1:1), 200°C, 24 h	5.2 ^a	—	374

^aObtained via X-ray photoelectron spectroscopy.

^bObtained via elemental analysis.

techniques with the milestones achieved within the field. It is important to note that thermal reduction has been proven to give the best quality rGO with highest structural quality, as indicated by the high conductivity obtained in contrast to other techniques as shown in Table 2. It is also noted that it is possible to achieve a thorough conversion of graphene oxide into graphene sheets with heat treatment of above 1773 K.⁹⁰

This review focuses on liquid-phase methods that produce graphene without strong oxidation effects. Other reviews from Park and Ruoff,⁹² Gengler et al.,⁹³ and Thakur and Karak⁹⁴ provide further details on the chemical exfoliation route and environmentally friendly graphene oxide reduction methods.

Electrochemical exfoliation

Unlike methods that rely on harsh solvents or oxidants, electrochemical methods take advantage of graphite's conductive properties to intercalate molecules between the graphene layers and therefore trigger exfoliation. Using graphite as an electrode, a positive or negative charge can be imposed on the material, promoting intercalation of oppositely charged ions and facilitating exfoliation. The mechanism for this process is illustrated in Figure 11.⁹⁵ Oppositely charged species enter between the sheets, expanding them, and reducing the exfoliation energy. Through application of either mechanical based

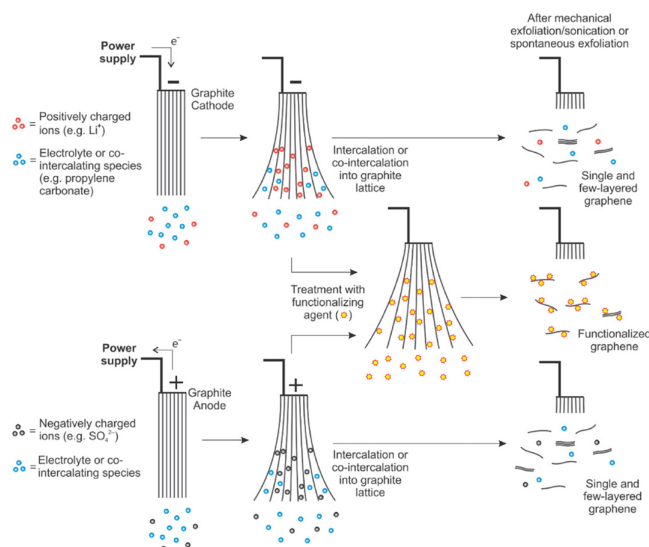


Figure 11. Schematic overview of cathodic and anodic exfoliation mechanisms.

A positive or negative charge is created at a graphite working electrode, attracting oppositely charged intercalating ions. Reproduced with permission from Ref. 95. Copyright 2015 Elsevier B.V. [Color figure can be viewed at wileyonlinelibrary.com]

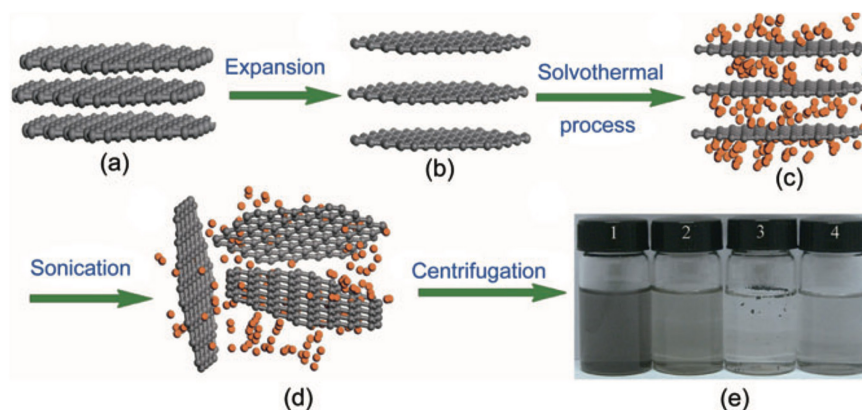


Figure 12. Schematic illustration of solvothermal-assisted exfoliation and dispersion of graphene sheets in acetonitrile: a) pristine expandable graphite. b) expanded graphite. c) schematic showing the insertion of acetonitrile molecules into the interlayers of expanded graphite. d) exfoliated graphene sheets dispersed in acetonitrile. e) optical images of four samples obtained under the different conditions described in Table 3.

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processes or sonication, the sheets can be separated in solution.

Exfoliation Methods

The following methods have demonstrated liquid-phase exfoliation of graphite into graphene with minimal or no adverse oxidative effects. The number of novel top-down methods is continually growing and, therefore, the list presented below is not intended to be exhaustive. It contains, however, a diverse range of common approaches that are relevant within the chemical engineering community. A brief summary of each approach is laid out in this section.

Solvothermal route

High-quality graphene can be produced at high-yield through a solvothermal process. This is a relatively simple, few-step process, to produce monolayer and few-layer graphene. Graphite flakes are dispersed in an organic solvent, acid, or other suitable solvent that could effectively penetrate through the interlayers and heated at elevated temperatures (from several hundred degrees Celsius to a range of 1000–2000°C).

Qian et al.¹⁶ produced monolayer and bilayer graphene by solvothermal-assisted exfoliation. In the first process step, expandable graphite was heated to 1000°C under a controlled atmosphere (95% of Ar, 5% of H₂) to enlarge the spacing between graphene sheets (Figures 12a, b). The expanded graphite was then mixed with a highly polar organic solvent, acetonitrile, in a Teflon-lined autoclave (without adding any surfactants or stabilizers). The mixture was heated for a period of 12 h at 180°C (Figure 12c). During this process, the

acetonitrile molecules gained sufficient energy to overcome the potential barrier present in the expanded graphite interlayers and diffuse into them. After which the mixture was sonicated, resulting in monolayer and bilayer graphene sheets (Figure 12d). Finally, centrifugation was used to obtain monolayer and bilayer graphene with a yield of 10 to 12 wt % at 600 rpm (Figure 12e). Separation of monolayer and bilayer graphene from expanded graphite was achieved by tuning the centrifugation velocity. The mixtures labeled by 1 to 4 in Figure 12e correspond to samples obtained using the process and centrifugation conditions stated in Table 3. More intense centrifugation enhanced the proportion of monolayers. This was accompanied by a reduction of the yield from ~12 to ~1 wt %. The lateral size of the graphene sheets was shown to be widely distributed, ranging between 100 and 800 nm. Insignificant structural defects, such as scrolling and folding, were observed in the bilayer product. Otherwise, electron diffraction and Raman spectroscopy confirmed that the graphene produced was defect-free.

Al-Hazmi et al.⁹⁶ produced graphene using a solvothermal process by mixing graphite flakes with tartaric acid followed by a short (60 s) thermal shock at 750°C to expand the graphite. This mixture was hydrothermally treated with isopropanol at 400°C for 120 min in a Teflon-lined autoclave. The autoclave was then quenched in an ice bath, and the product was appropriately filtered and left to dry overnight at 60°C. Raman spectroscopy revealed that graphite was successfully expanded after the thermal shock, and subsequently, graphene sheets with low defects and oxidation degree (D to G intensity ratio of 0.06) were produced. The 2D/G intensity ratio was found to be around 2.3, indicating the presence of monolayer graphene

Table 3. Production Metrics (or yield) for Different Types of Graphite and Exfoliation Conditions. Reproduced with permission from Ref. 16. Copyright 2009 Tsinghua University Press and Springer Berlin Heidelberg

Sample Number (see Figure 12e)	Carbon Source	Process	Centrifugation Conditions	Yield (wt %)
1	Expanded graphite	Solvothermal-assisted process	600 rpm, 90 min	10–12
2	Expanded graphite	Solvothermal-assisted process	2000 rpm, 90 min	0.8–1
3	Expanded graphite	Without solvothermal-assisted process	600 rpm, 90 min	None
4	Sieved graphite powder	Without solvothermal-assisted process	600 rpm, 90 min	0.5–0.6

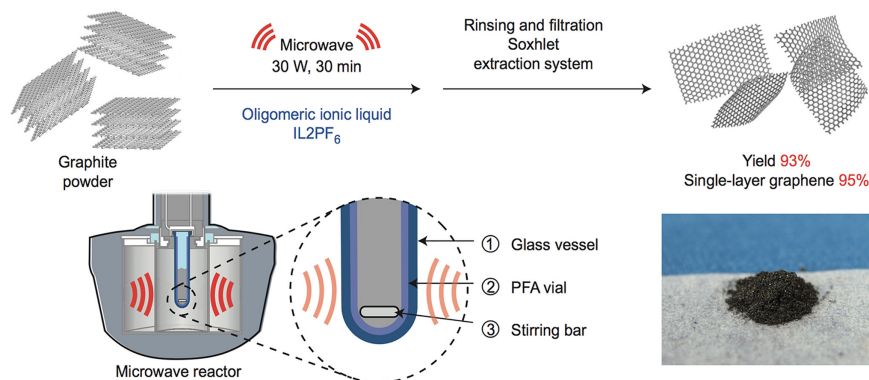


Figure 13. Overview of the microwave-assisted liquid-phase exfoliation of graphite in ionic liquids with a schematic of the “reaction” setup.

Typically, a vial containing a graphite suspension in ionic liquid was placed in a glass reaction vessel. This vessel was then placed in a microwave reactor operating at 2.45 GHz and irradiated using a single-mode microwave setting at 30 W for 30 min. The microwave-irradiated mixture of graphite in ionic liquid was diluted with dimethylsulfoxide (DMSO), and the entire mixture was transferred to a PTFE thimble filter. The solid residue in the thimble filter was rinsed successively with DMSO (reduced pressure; ~30 mmHg), 2,2,2-trifluoroethanol and dichloromethane using a Soxhlet extraction system, and subsequently dried under reduced pressure to create a black powder (bottom right) that contained 95% single-layer graphene (93% yield). Reproduced with permission from Ref. 27. Copyright 2015 Nature Publishing Group. [Color figure can be viewed at wileyonlinelibrary.com]

sheets. Using 100 graphene sheets as a sample size, it was estimated that ~95% of the graphene sheets had an average thickness of 0.53 ± 0.04 nm (monolayer), while ~5% had an average thickness of 0.75 ± 0.05 nm (bilayers). The lateral dimensions were found to be distributed between 250 and 450 nm. Finally, X-ray photoelectron spectroscopy (XPS) showed that the final graphene sheets contained ~6% oxygen.

Liu et al.⁴² prepared graphene sheets with 2–6 layer thicknesses in an ethanol solution by the solvothermal route, and reported several important findings. First, increasing temperature in this process removes oxygen-containing groups, and better quality graphene sheets are obtained by essentially restoring π -bonding. Second, the D to G intensity ratio increases as the solvothermal temperature rises. This could be explained by the formation of defects on removal of oxygen atoms at increasing temperatures. TEM characterisation showed previously observed structural defects such as curled and partially folded structures. HRTEM revealed the presence of graphene sheets with various thicknesses that are in the range of 2 to 6 layers.

Further information can be found in a review by Sasikala et al.⁹⁷ which takes an in depth view of solvothermal processing, including how the synthesis condition affects the morphology and band properties of the materials obtained.

Microwave irradiation

Microwave irradiation is another example of thermal exfoliation of graphite. This approach is typically faster than the solvothermal route and has been reported in a number of papers. For instance, microwave irradiation in ionic liquids was investigated by Matsumoto et al. Using this method, graphite was dispersed in an ionic liquid and irradiated with microwaves for 30 min (Figure 13). The setup is somewhat similar to ultrasonication, but the processing time is greatly reduced, and an improved route for graphene exfoliation was demonstrated. The authors reported a high exfoliation efficiency, a yield of 93 wt % and a selectivity of 95% toward single-layer graphene.²⁷ The D/G ratio measured by Raman showed a value of ~0.14 (similar to the precursor graphite with value of ~0.13), demonstrating a high-quality graphene product. The authors

suggested that HF species, generated under microwave irradiation by partial decomposition of the fluorine-containing ionic liquids, play a crucial role in the exfoliation. Sterically less-demanding HF molecules first intercalate stochastically into the thermally agitated interlayers of graphite. This produces structural disordering of the graphite, and promotes intercalation of the sterically more-demanding ionic liquid molecules with enhanced affinities. This effect is attributed to the splitting of graphitic layers.²⁷

Liu et al.⁹⁸ demonstrated a fabrication method of multilayer graphene flakes produced with an increased exfoliation efficiency through intercalation and decomposition of H₂O₂ and (NH₄)₂S₂O₈. Starting with commercially available expandable graphite, first, they have processed it for 10 s using microwave heating before mixing and intercalation individually with the above-mentioned compounds. It is reported that intercalation begins to occur at the defect and edge sites as well as grain boundaries. Once these compounds start to decompose into various gases e.g. oxygen and steam, this leads to an increase in the graphite interlayer distance allowing even more molecules to participate in the intercalation. Subsequently, this results in an effective expansion and exfoliation of graphite in the process of second microwave heating. This mixture is then sonicated and centrifuged to obtain a stable dispersion of graphene sheets, which was then used for characterisation. AFM images showed that the mean thickness of multilayer graphene sheets prepared in H₂O₂ was about 6.15 nm with lateral sizes reaching several micrometers, whereas multilayer graphene sheets prepared in (NH₄)₂S₂O₈ had larger mean thickness of approximately 9.15 nm. Raman spectroscopy have showed that values for both the I_D/I_G and I_D/I_{2D} are noticeably lower for each mixture compared with precursor and initially heated graphite materials. This demonstrates that further heating by microwaves allows to restore crystalline quality. FTIR spectra showed that graphene product prepared in H₂O₂ has much less oxygen-containing functional groups than the graphene product prepared in (NH₄)₂S₂O₈. This may occur due to the fact that oxide radicals are released as ammonium persulfate decomposes, which initiate radical induced cutting along the edges and surface defects.⁹⁹ Similarly, Liu et al.¹⁰⁰ have

fabricated high-quality graphene sheets which had lateral size of up to 100 μm and thickness of just few layers. Comparing to the conventional Hummer's method,¹⁰¹ this research group claims that their prepared graphene sheets are on average 10 times larger in size. Moreover, in their graphene fabrication process, no postprocessing such as ultrasound, centrifugation, shock or H_2O_2 treatments are required. They used commercial expanded graphite and applied microwave irradiation for 10 s, following which the volume of this graphite powder expanded dramatically. It was then mixed with concentrated sulfuric acid and KMnO_4 , where the reactant ratio of KMnO_4 to expanded graphite was set to 2:1. The whole mixture was then further heated by microwave radiation for a further 10 s. This processed powder was characterized using SEM and TEM which proved to contain transparent, continuous, wrinkled, and agglomerated microstructures of graphene sheets. As characterized by AFM, prepared graphene-oxide sheets had lateral dimensions of about 50 μm and mean thickness of approximately 2.15 nm (but ranging between 1 and 12 nm). Analysis by XPS showed that the produced graphene oxide sheets have C/O ratio of 2.6. Structural properties were probed using Raman spectroscopy, which confirmed the intensity ratio between D and G bands (I_D/I_G) to be 0.85. Oxygen-containing functional groups were identified using FTIR, and the results clearly showed the effectiveness of deoxygenation/reduction by the final microwave heating step. Comparing oxidized and intercalated graphite powder with the final microwaved graphene sheets, it is evident from the spectra that the latter has much weaker peaks than both the former and precursor materials.

Another interesting work was conducted by Lin et al.,¹⁰² where the group used ammonium bicarbonate as the intercalation product (without the addition of solvents) in the microwave-assisted exfoliation of graphite. This graphene production method, which involved pristine graphite and ammonium bicarbonate agitation, thermal treatment and microwave irradiation, produced 5–8 layers graphene sheets with lateral sizes of 0.4–0.8 μm as characterized using AFM. Evidenced by TEM, the edges of the produced graphene sheets were partially folded and had wrinkled surfaces. Moreover, the presence of graphene in the final product was confirmed by XRD and Raman spectra. Analysis performed using XPS revealed that the O/C ratio was 0.168, meaning that only several oxygen-containing groups were attached to graphene sheets. According to Tang et al.,¹⁰³ ammonium bicarbonate acts as an effective dispersant in graphite powder and prevents graphene from agglomeration. It also attaches to the edge defects of graphite by heating and cooling treatment of this mixture. Thermal decomposition of ammonium bicarbonate leads to formation of various gaseous species (e.g., CO_2 , NH_3 , H_2O) which can overcome graphite interlayer van der Waals forces and subsequently, results in effective exfoliation of graphite.

Supercritical fluids

The use of supercritical fluids has also been extended to exfoliating graphene sheets. The critical factors to consider using this technique are the high diffusivity, expansibility, and solvating power of the supercritical fluid. The supercritical fluid can penetrate the gap between the graphite layers owing to the high diffusivity and low viscosity of the supercritical fluid. Once a rapid depressurisation happens, the supercritical fluid will abruptly expand to predominately generate a normal force for exfoliating graphite as seen in Figure 14a.

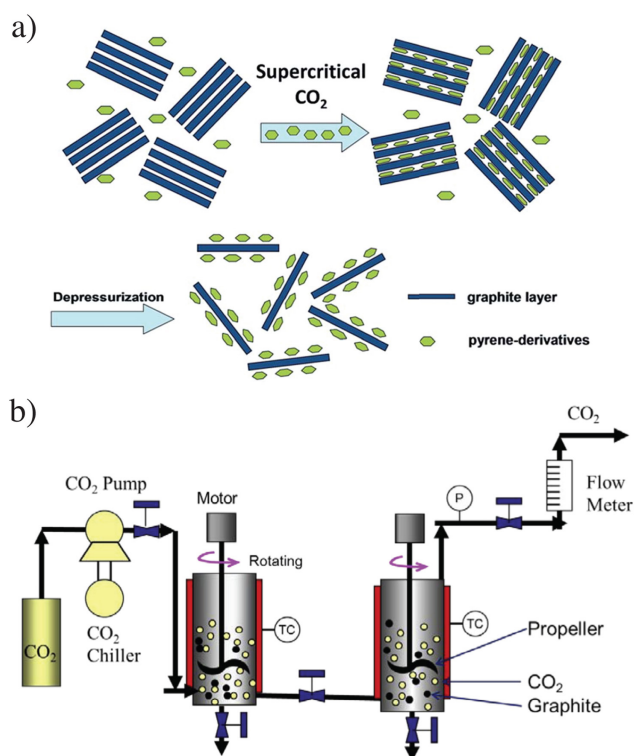


Figure 14. a) Schematic of the exfoliation of graphite using a supercritical fluid (i.e. CO_2). In this instance, graphene was simultaneously modified with pyrene-derivatives. b) Schematic of an experimental device to exfoliate graphite using a supercritical fluid.

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Pu et al. used supercritical CO_2 gas to exfoliate graphite in a solution containing a sodium dodecyl sulphate dispersant. Graphene flakes of typically 10 atomic layers in thickness were obtained.⁴³ In a different study, Rangappa et al. used supercritical ethanol, NMP and DMF for the exfoliation.¹⁰⁴ The solvents were heated up to, or above, their critical temperature and the exfoliation of few-layer graphite (<10 layers) was achieved in the shortest reaction time of 15 min. Approximately 90–95% of the exfoliated sheets were less than 8 layers with approximately 6–10% monolayers.¹⁰⁴

An attempt to scale-up graphene production using shear-assisted supercritical CO_2 exfoliation was reported by Li et al.³⁹ The experimental apparatus is shown in Figure 14b. In all experiments, the process time was fixed at 1 h. When CO_2 fluid with graphite powder is subjected to a high shear stress, graphite flakes expand and delaminate into multilayer graphene sheets. Temperature was found to be an important factor as it directly affects the graphene yield. An increase in temperature from 35 to 55°C, subsequently increased graphene yield from 40 to 87%, when other parameters were kept constant. This is due to the enhanced energy kinetics of the CO_2 molecules and, therefore, thermal motion. It was also observed that higher pressure slowly increased exfoliation yield. This trend is related to the larger quantity of CO_2 molecules at high pressure that can penetrate the interlayer space of graphite. The effect of the mass of graphite on the exfoliation process was

also investigated. The study showed that increasing the amount of graphite by 95 wt % lowered the exfoliation yield from 80 to 40%. This is due to an insufficient CO_2 /graphite ratio causing smaller shear stress. Rotational speed was another parameter investigated. It was found that increasing rotor speed from 0 to 2000 rpm, increases the exfoliation yield from 10 to 70%. The main driver of this is an increase in turbulent shear stresses in the mixture, which overcomes the van der Waals force barrier with a greater effect than at lower rotational speeds and they were able to obtain 90% exfoliated sheets with less than 10 layers, and 70% between 5 and 8 layers.³⁹

Recently, the combination of supercritical fluid and sonication was reported by Gao et al.²⁰ In this study, graphene flake distributions were 24% monolayers, 44% bilayers, and 26% trilayers. Song et al.¹⁰⁵ investigated a fluid dynamics method in supercritical CO_2 . Scale-up of graphene production was attempted by combining two exfoliation mechanisms to increase the yield. A shear method, coupled with the expanding CO_2 exfoliation method, was implemented. This approach led to a yield of $\sim 63\%$ with a selectivity of monolayer graphene of 27%. Overall, a major challenge of the supercritical fluid method is the need for pressurized reactors to facilitate the process. For example, the work of Gao et al.²⁰ required pressures of 12 MPa to achieve the above stated yields.

Vortex fluidic device

Another shear-driven process to exfoliate graphite in an organic solvent was developed by Chen et al. This approach used a rapidly rotating tube to generate an intensive shearing environment for the production of graphene from graphite.¹⁰⁶ A schematic of the setup is shown in Figure 15. A thin liquid (i.e., NMP) film was created when a glass tube was set to rapidly rotate at an angle varied between 0° and 90° . High internal shear stresses within the thin liquid film provided the necessary conditions for exfoliation. The balance between gravitational and centrifugal forces was adjusted, by varying the previous parameters, to examine the impact on exfoliation performance. A rotational speed of 7000 rpm, an inclination angle of 45° , and a processing time of 30 min were found to be the optimal settings for graphene exfoliation. TEM characterisation established the yield of the graphene sheets to be ≤ 1 wt %. The yield of monolayer graphene sheets depended on the morphology of the precursor graphite flakes (i.e., size, chemical, and physical properties of the edges), and the orientation and magnitude of the shear force. Selected Area Electron Diffraction (SAED) confirmed a hexagonal distribution of peaks for graphene, and AFM height profile measurements were approximately 1 nm.

It was proposed that Stewartson/Ekman layers are formed by liquid flowing downwards close to the gas–liquid interface, and liquid flowing upwards close to the wall of the rotating tube (Figure 15c). In the rapidly rotating fluid, the shear layers are parallel to the axis of rotation. The large centrifugal force initially accelerates graphite flakes to the walls of the tube. For a tube rotating at 7000 rpm and inclined at an angle of 45° , this produced an average liquid film thickness of approximately $230\text{ }\mu\text{m}$. The average thickness of the flakes was less than $1\text{ }\mu\text{m}$. Chen et al. suggested that the flow rate above the flakes is close to the maximum.¹⁰⁶ The mechanisms shown in Figures 15d, e for graphite layer-by-layer exfoliation in a thin film were supported by the partially stacked graphene sheets observed using TEM. The exfoliation involved a slippage

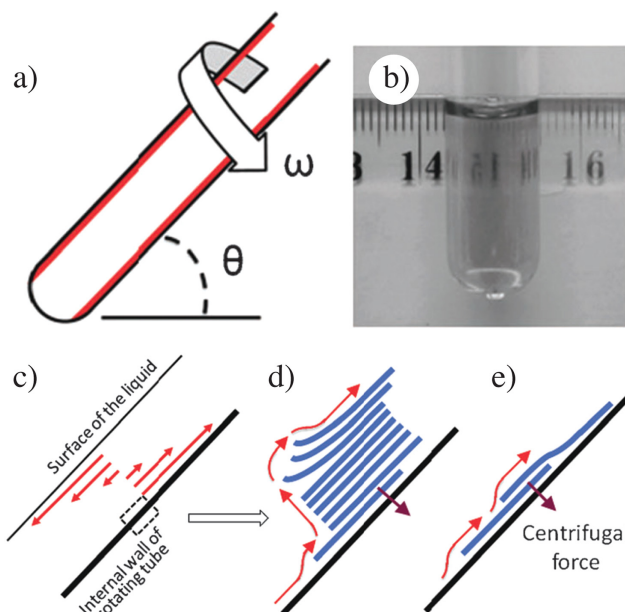


Figure 15. a) Schematic of the vortex fluidic device used to exfoliate graphite and boron nitride (10 mm diameter tube, 16 cm long, inclined at 45° , operating at 7000 and 8000 rpm for graphite and boron nitride, respectively). b) Photographs of the resulting colloidal suspensions of graphene (top) in NMP. Illustrations of c) the microfluidic flow velocity indicated by red arrows for a section of the rotating tube; d) the exfoliation process; and e) slippage on the inner surface of the tube.

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process, where the graphene sheets are displaced relative to each other. This requires lifting individual sheets from the surface of the bulk graphite through the presence of lateral force. Graphene sheets were found to be attached to the walls of the tube, suggesting that this slippage of graphene sheets can also occur on the surface of the tube. A large centrifugal force can, therefore, hold the graphite flakes against the internal tube wall and simultaneously provide a shear-induced displacement along the surface.

Wahid et al.¹⁰⁷ used a similar vortex fluidic device, and experimental parameters to those developed by Chen et al.¹⁰⁶ in an attempt to produce a graphene hybridized material with algal cells for use in the removal of nitrate from liquid effluents. First, graphite was dispersed in water and exfoliated into multilayer graphene using a vortex fluidic device at the optimal operating conditions that created the required environment for intense shearing. Materials characterisation using SEM, TEM, and AFM indicated the presence of multilayer graphene sheets. The multilayer sheets had varied thicknesses, some thicker than 20 nm and a small portion of ~ 1 nm thick flakes. The characteristic peaks from Raman spectroscopy indicated that the graphene sheets were ≥ 5 layers thick and contained similar defects to those present in precursor graphite flakes. Hence, the results obtained from the different characterisation techniques imply that the vortex fluidic device can only produce multilayer graphene when operated at the optimal

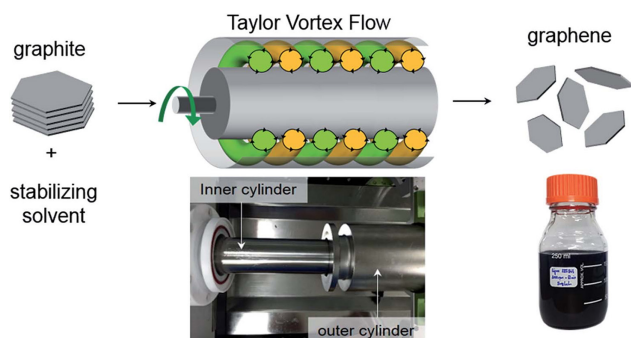


Figure 16. Schematic of the shear-exfoliation of graphite into few-layer graphene by a Taylor Vortex flow.

Photographs shows a Taylor–Couette flow reactor and the graphene dispersions produced by shear exfoliation. The volume in the reactor for mixing was 200 mL, and the gap between the two concentric cylinders was 2.5 mm. Reproduced with permission from Ref. 47. Copyright 2016 The Royal Society of Chemistry. [Color figure can be viewed at wileyonlinelibrary.com]

parameters (i.e. 7000 rpm and 30 min processing time). It may be possible to thin-out these layers closer to monolayer sheets if processing time was prolonged. This, however, would offer a less practical approach towards mass-scale graphene production. The approach is less energy intensive than sonication and offers a more feasible approach for the scale-up graphene production.

Taylor-Couette device

Tran et al.¹⁰⁸ demonstrated that a high shear environment, created in a Taylor-Couette flow reactor, could be used to exfoliate graphite flakes with a relatively high yield. Graphite flakes, dispersed in a stabilising solvent (NMP), were processed using this method. The liquid suspension flows through a small gap between a rapidly rotating inner cylinder and stationary outer cylinder (Figure 16). Once the inner cylinder is rotating above a critical rotational speed, a secondary flow occurs which is distinguished by the presence of counter-rotating toroidal vortices, also known as Taylor vortex flow. A sufficient local shear stress at the reactor wall was created, leading to the exfoliation of graphite flakes into graphene sheets with a low degree of defects.

The graphite/solvent dispersion was continuously recirculated through the reactor and a separate chiller (to prevent heating of the mixture) for 10 to 120 min. Rotation of the inner cylinder was varied from 500 to 3000 rpm. The mixture was centrifuged and the exfoliated graphene in the supernatant was collected for characterisation. SEM images were used to observe graphene sheets which were thinner than the precursor graphite flakes. Variation in flake lateral sizes were between 500 and 1500 nm (as observed from TEM images). This size distribution potentially indicates that a fragmentation mechanism occurred in the exfoliation process. This could also be due to the wide particle-size distribution of the graphite precursor. An electron diffraction pattern confirmed the presence of graphene monolayers. AFM images confirmed the values of the flake size and showed that most of the flakes were less than 3 nm thick, corresponding to few-layer graphene sheets. The thinnest sheets observed were 0.6 nm. A sample size of 250 flakes were also characterized by AFM to determine the height profile. This showed that over 90% of these flakes were

less than five layers thick. A low degree of defects was evident with an I_D/I_G ratio of ~ 0.14 . Furthermore, XPS showed that produced graphene sheets were not oxidized, confirming the high quality of these flakes.

Spinning disc processor

Using a spinning disc processor, Chen et al. generated a shear force in dynamic thin films across a disc surface. This led to the exfoliation of graphite flakes, and production of carbon nanoscrolls.¹⁰⁸ Hexagonal boron nitride (h-BN) scrolls were also produced using this technique. A schematic diagram of the spinning disc processor is shown in Figure 17a. A jet feed of the suspension (i.e., graphite or h-BN flakes dispersed in NMP) is directed near the centre of a 20 cm diameter disc that has a grooved surface. Due to the high-speed of rotation (2500 rpm), the liquid spreads across the disc forming a thin film with high surface area-to-volume ratio. This speed results in a short residence time for the dispersed particles (< 1 s as noted in this paper). Hence, the 200 mL solution is recycled for over 20 h at 60 mL min^{-1} flow rate.

From the TEM images, carbon nanoscrolls (with approximate yield of 1%) were both multilayered (exact nanoscroll thickness was not specified) and irregular in shape. Various sheet deformations were observed, shown in Figure 17c). These deformations may have formed either due to high-velocity impact or the intense shearing environment. The impact occurs when particles leave the surface of the rotating disc, and strike the walls of the tank surrounding the disc. The irregular material shapes vary considerably in length and diameter, which could be due to a variation in the original graphite flake sizes. A significant number of partially formed

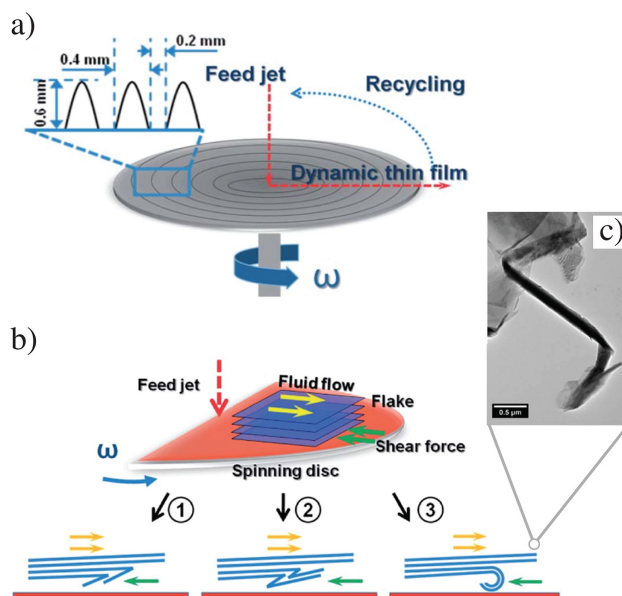


Figure 17. a) Schematic of the spinning disc processor used for exfoliation of graphite (20 cm diameter stainless steel disc operating at 2500 rpm). b) Proposed mechanism of exfoliation and scrolling of graphite and boron nitride flakes using the setup. c) TEM image of partial carbon nanoscroll.

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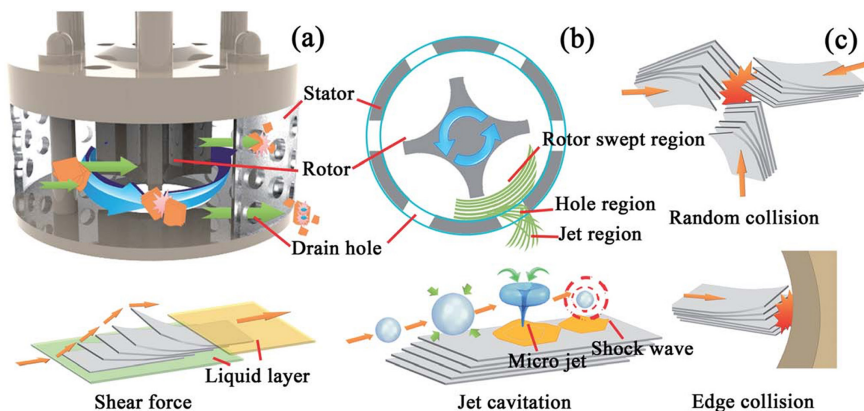


Figure 18. a) 3-D sectional drawing of a high-shear mixer for the production of graphene. b) Main energy dissipation regions of the high-shear mixer (sectional view). c) Schematic of preparing graphene nanosheets by shear force, collision and jet cavitation.

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carbon nanoscrolls were also observed. This could be caused by the irregular morphology of the graphite flakes under varying localized microfluidic conditions. Chen et al. proposed a potential mechanism, illustrated in Figure 17b. Particles, suspended in NMP, are immediately accelerated by the centrifugal force when introduced from the feed jet onto the surface of the spinning disc. The liquid suspension quickly spreads across the disc, forming a dynamic thin film typically $\leq 200 \mu\text{m}$ thick.¹⁰⁸ Suspended flakes move throughout the thin film, with periodic contact with the surface of the rotating disc. High shear forces occur from the viscous drag of the liquid on the surface of the disc and result in exfoliation. Grooves through entire surface of the disc, shown in Figure 17a, may further enhance the frequency of the collisions of the flakes with the surface of the disc.

Analysing SEM images, the authors proposed three feasible scenarios for exfoliation occurring due to shearing (Figure 17b). The first two mechanisms may involve exfoliation of the bottom layers with different degrees of folding due to lift from the shear force on the upper layer of the flakes. The third mechanism may involve curving of the flake edges to form scrolls. As these three mechanisms may compete with one another, their occurrence will depend on various factors such as flake orientation, nature of the material, number of layers exfoliated at once, shear intensity, and so forth. Molecular simulations for a single graphene sheet were also performed, and an analytical continuum model for the scrolling of the graphite flake provided an insight on the scroll formation.

High-shear mixing

High-shear mixing has been demonstrated in the literature as a potential method for scale-up of graphene production by Paton et al.³⁵ Liu et al. used a high-shear mixer to exfoliate graphite and produce graphene flakes. The head of this mixer contained a rotor and a stator with an adjustable gap of between 100 and 3000 μm (Figures 18a, b). When the rotor operated at high speed (10–50 m s^{-1}), high shear rates (20,000–100,000 s^{-1}) forced the suspension to flow through the narrow gaps between a rotor and a stator.³⁶

Liu et al. identified that the graphene sheet size and thickness can be controlled by setting a specific gap size between rotor and stator.³⁶ In fact, they have observed that a small stator produces smaller graphene sheets than the larger stator.

Approximately 1000 flakes were analysed for average size and thickness to assess the effect of the gap size. Using a small stator, the graphene sheets had an average size and thickness of 0.35 μm and 0.9 nm, respectively. Those prepared by a large stator had an average size and thickness of 0.9 μm and 1.3 nm, respectively. Hence, higher shearing stresses (achieved using small stator with narrower gaps between rotor and stator) are preferred to produce thinner and smaller graphene flakes relative to a larger stator with wider flow channels.

Similarly, the mixing effect on graphite exfoliation was also explored by Paton et al., using a mixer with a rotor/stator head. TEM was used to measure the size of the nanosheets, found to be in the range of 300 to 800 nm. AFM images showed that the graphene sheets consisted of 10 layers and less. Raman spectroscopy confirmed the presence of graphene monolayers. The authors suggested that turbulent energy dissipation is not necessary to achieve effective graphite exfoliation. Production characteristics were assessed for varying rotor speed and rotor diameter. As shown in Figure 19a, graphene was produced at both low ($\text{Re} < 10^4$) and high Reynolds number ($\text{Re} > 10^4$). Reynolds number was defined as $\frac{ND^2\rho}{\eta}$, where N is rotor speed, D is rotor diameter, ρ and η are the liquid density and viscosity, respectively. Graphene was produced at a minimum shear rate of 10^4 s^{-1} (Figure 19a) and within a laminar flow regime, supporting the suggestion that turbulent flow is unnecessary for exfoliation. Several possible mechanisms for graphite exfoliation were offered and are illustrated in Figure 18c). These include shear force, cavitation, and various types of collisions. Although the shear mixing method has demonstrated some of the highest production rates currently (i.e., 5.3 g/h), there may be limitations in yield and throughput, as the high shear rates are generated locally within a micron-scale rotor-stator gap.

Ultimately, it can be deduced that the same exfoliation mechanism occurs in both turbulent and laminar flow regimes. Any mixer, therefore, that can generate a shear rate of 10^4 s^{-1} and above, should be capable of producing graphene. Paton et al. demonstrated this using a kitchen blender, where local turbulent shear rates of more than 10^4 s^{-1} are generated by the rotating blade.

A more detailed work using a kitchen blender was performed by Yi et al.¹⁰⁹ to understand the viability of this process in the production of graphene. The process used rapidly

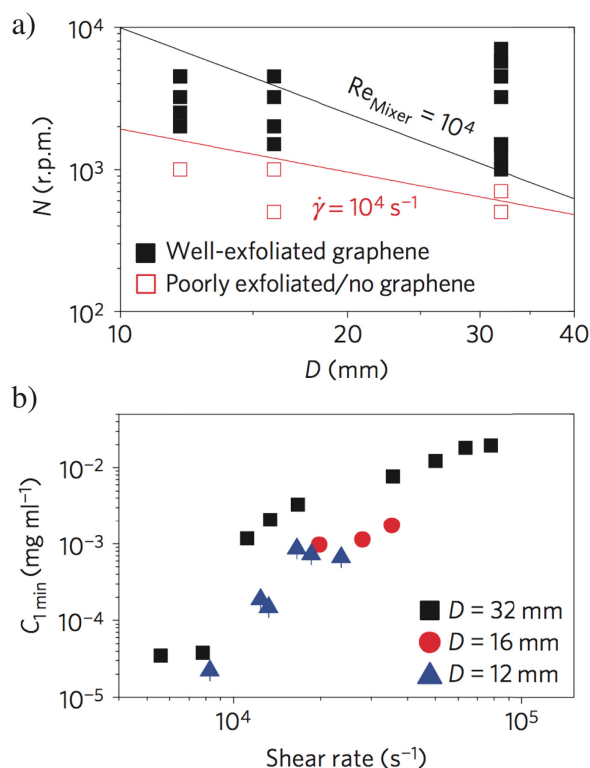


Figure 19. Overview of the outputs using a high shear mixing process for exfoliation.

a) Phase diagram of rotor speed, N , versus diameter, D , for dispersions showing good exfoliation according to TEM. The region above the black line represents fully developed turbulence, that is, $Re_{Mixer} > 10^4$, whereas the region above the red line represents $\dot{\gamma}_{min} > 10^4 \text{ s}^{-1}$. b) Concentration of graphene produced in the Silverson shear mixer as a function of shear rate for rotors with diameters of 32, 16, and 12 mm (mixing time 1 min). All three data sets are consistent with the same minimum shear rate. Reproduced with permission from Ref. 35. Copyright 2014 Nature Publishing Group. [Color figure can be viewed at wileyonlinelibrary.com]

rotating mixer blades that create a high-shear region across a larger volume than that in the rotor/stator space. Although this high shear region is not as localized, the shear rate decreases with increasing distance from the mixer blades. Hence, there is an opportunity to develop a high shear region across the whole mixing volume if suitable engineering design and optimal process parameters are applied. Once this is established, there are four exfoliation mechanisms that could occur within this region: (1) viscous shear stress induced by a velocity gradient, (2) Reynolds shear stress induced by turbulent velocity fluctuations, (3) graphite-graphite collisions due to high inertial forces that dominate viscous forces in turbulent regions, (4) normal-forces caused by pressure differences that arise from turbulent pressure fluctuations.

Varrla et al.¹¹¹ extended the use of the kitchen blender to produce graphene by exfoliating graphite dispersed in a surfactant. AFM and TEM characterisation results showed that mono- and few-layer graphene sheets were successfully produced but contained some folded flakes. For this specific model of kitchen blender, an optimal blade rotational speed was identified to be around 2000 rpm. This was found by analysing graphene concentration produced under a variable blade rotational speed.

Both Yi et al. and Varrla et al. presented graphene flake size and thickness distribution using AFM results. Yi et al. set the rotational speed of the blade to 5000 rpm and performed the experiment for 8 h, collecting samples at regular time periods. Approximately 92% of flakes analysed were less than 1.5 nm thick. The fraction of flakes with ≤ 1 nm thickness was between 14.6 and 20% for all mixing times. It was also observed that longer processing times result in noticeable decreases in flake area (i.e., from $\sim 2.4 \mu\text{m}^2$ at 30 min to $\sim 0.1 \mu\text{m}^2$ at 8 h). These values can be compared with those obtained by Varrla et al., where the rotation speed of 18000 rpm for 1 h was used. The average sheet length and average layer number were 320 nm and 6, respectively.

Sonication

Sonication is a liquid-phase exfoliation technique that has been widely employed to produce graphene monolayer and multilayer sheets. The process typically involves a subsequent centrifugation step as a way to separate the residual graphite flakes from the exfoliated sheets. Sonication can be used either solely, or be combined with other techniques.

Exfoliation via sonication relies on the production of high intensity ultrasonic waves that travel through the liquid media at a specific frequency. Cavitation bubbles are produced, and grow to the order of microns in diameter. These bubbles become unstable and rapidly collapse during a high-pressure cycle as illustrated in Figure 20. These collapsing bubbles create hot spots where temperatures of $\sim 5000 \text{ K}$, pressures of ~ 1000 atmospheres, and heating and cooling rates of more than 1000 K s^{-1} are reached.¹¹¹ These cavitation-induced bubbles distribute around the graphite flakes and when bubbles collapse, micro jets, and shock waves can act on the graphite surface, producing compressive stress waves that propagate throughout the graphite body. From the theory of stress waves, once the compressive wave spreads to the free interface of graphite, a tensile stress wave will be reflected back to the body.

The collapse of numerous micro-bubbles will, therefore, lead to intensive tensile stresses in the graphite flakes. Two secondary processes are also possible. Unbalanced lateral

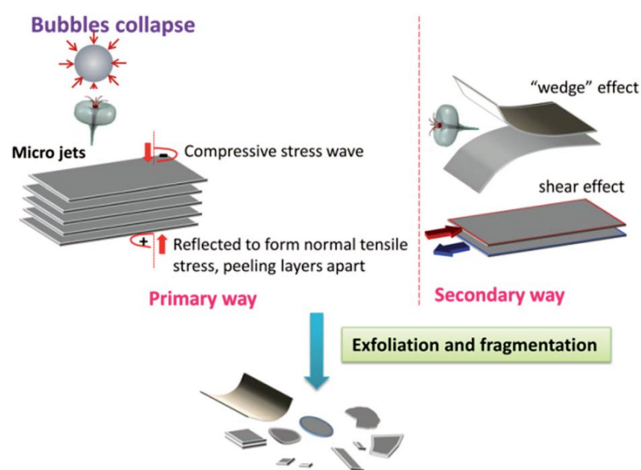


Figure 20. An illustration of the mechanical mechanism for exfoliation via sonication.

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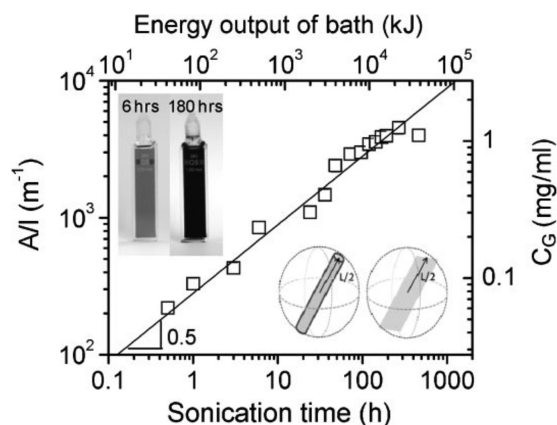


Figure 21. Concentration of graphene after centrifugation as a function of sonication time.

On the left axis is the measured absorbance per cell length, A/l , while the right axis corresponds to the concentration calculated using an absorption coefficient of $3620 \text{ mL mg}^{-1} \text{ m}^{-1}$. The upper axis shows the total energy output by the bath calculated using the measured power output of 23 W. Reproduced with permission from Ref. 23. Copyright 2010 John Wiley & Sons, Inc.

compressive stresses can separate two adjacent flakes: (1) by a shear effect, and (2) microjets may split the graphite flakes just as a wedge is driven into the interlayer. All this implies that it is the tensile stress that effectively exfoliates graphite into graphene flakes, resulting in a normal force dominated mechanism.⁸⁵

Hernandez et al.²¹ first reported high yield production of graphene via sonication. Graphite powder was dispersed in organic solvents such as DMF and NMP, followed by sonication and centrifugation. A graphene dispersion was obtained with a monolayer concentration of 28%. Although the set up and experimental method are relatively straight forward, other recent works showed that this method produces very low graphene concentrations, ranging from $\sim 0.01 \text{ mg/mL}$ to $\sim 1 \text{ mg/mL}$.²³ Prolonging sonication time to around 1000 h, higher concentrations of graphene in dispersions (Figure 21) were achieved, but this becomes impractical due to excessive processing time and energy.

Further work was done to achieve high concentration by adding surfactants and polymers, solvent exchange methods, and mixing solvents.^{40,41,45,48,51,74,112–116} All the methods within these studies are still plagued with the main drawback of low rate of graphene production and long processing time. In addition, the exposure of graphene particles to a high energy sonication environment also reduces flake size and increases the level of surface-edge defects.⁴⁰ Further development in this area is required in order to comprehend all the variables involved i.e. vessel geometry, position of the sonication probe, optimal sonication parameters under given conditions, solvents and surfactants safe for health and the environment, and how these can maximize graphene production with the minimum number of defects. Alternatively, more development works could be performed by combining ultrasonication with other graphene exfoliation methods, that is, fluid dynamics. We note three recent reviews by Lavin-Lopez et al., Ciesielski and Samori, and Muthoosamy and Manickam which expand into more detail on the topic of exfoliation via sonication.^{117–119}

Jet cavitation

Jet cavitation is another method that has been developed to capitalize on the effect of shearing. Liu et al.¹²⁰ found that jet cavitation can be induced with a high pressure difference in a cavitation generator (Figure 22a). Similar to the sonication method, the collapsing bubbles provide a mechanism for exfoliation of graphite into few layer graphene flakes. The system consists of multiple flow channels where the exfoliation occurs at various positions due to pressure differences between the inlet and the outlet. A continuous flow of graphite/solvent dispersion was injected to the pressure vessel and pressurized by a plunger pump. As illustrated in Figure 22b, this created many bubbles, an upon imploding, created microjet and shock-wave forces across the graphite surface. This mechanism produced graphene flakes, and is described in detail in the previous section. AFM showed that graphene flakes had a height range of approximately 0.8–0.9 nm. The HRTEM showed the presence of few-layer graphene sheets.

The exfoliation of graphite is also driven by other mechanisms, that is, turbulence-induced Reynolds shear stress, viscous shear stress, pressure release, and collisions of graphite particles with one another as well as channel walls, induced by turbulence. These mechanisms lead to lateral-, normal- and shear-force dominated exfoliation as well as fragmentation.

Liang et al.²⁶ produced graphene nanomesh (i.e., sheets with a porous structure) using a similar fluid-based device to that of Liu et al. An analysis of the flake distributions was conducted on 100 flakes collected from several AFM images. Over 70% of the flakes were 1 to 1.5 nm thick, with the remaining less than 1 nm thick. A minor amount were observed to be thicker than 1.5 nm. Liang et al. have illustrated several exfoliation mechanisms of this scenario. Liang et al.¹²¹ also highlighted that the variation of the treatment

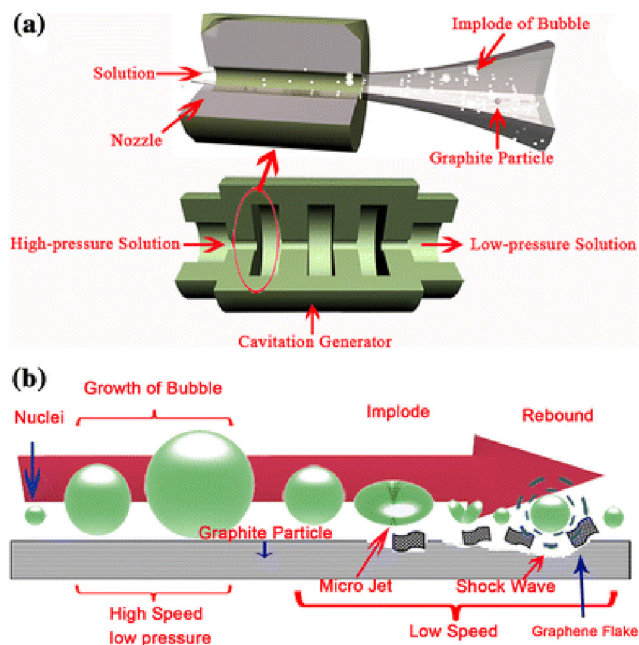


Figure 22. a) Schematic of the cavitation generator. b) Illustration of the mechanisms during the production of graphene flakes by jet cavitation.

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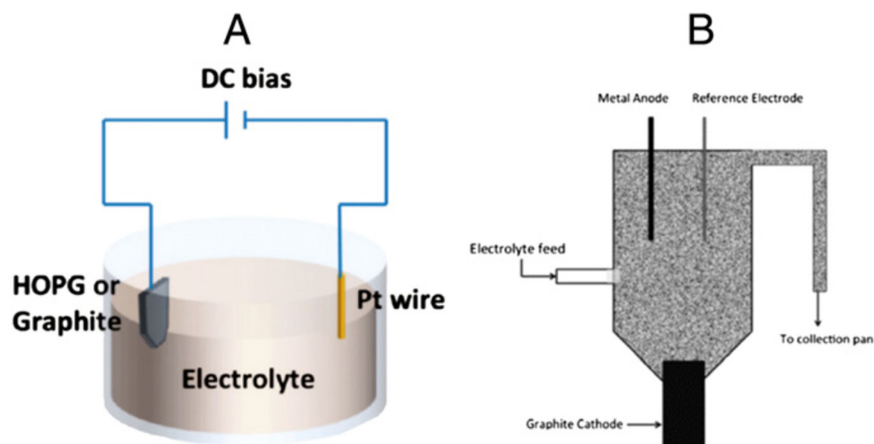


Figure 23. a) Schematic of a typical setup for the electrochemical exfoliation of graphite. b) A novel setup for continuous electrochemical exfoliation process.

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time and pressure directly affects thickness, lateral size and concentration of graphene by jet cavitation. As the treatment time is increased, there is a shift of thickness distribution to a lower and narrower range of values. The total amount of graphene flakes with a thickness of less than 1.5 nm increases from 29% after 30 min treatment, to 63% after 4 h treatment, and 79% after 8 h treatment. The same applies for the flake area, where the values drastically change from a wide distribution after 30 min treatment to a narrow distribution after 8 h treatment with 85% of the exfoliated flakes having $<10^5 \text{ nm}^2$. Prolonging treatment time and applying higher pressure also have a positive effect on the graphene yield.

Chaotic flow with sonication

Liu et al.³⁸ performed liquid exfoliation of graphite in NMP (initial concentration 6 mg mL^{-1}) by so called chaotic flow followed by sonication. This suspension was placed into a sample holder of the chaotic flow apparatus. Within the sample holder there were four cylindrical ceramic pebbles that lead to particle-pebble and graphite inter-particle collisions. Chaotic flow is created through rapid separation of liquid streams around these pebbles. Liu et al. proposed two mechanisms that most likely occur during this exfoliation process. The first one is the high-shear environment that is created between the suspension and the pebbles. The second mechanism may occur due to the formation of curled-up edges of the graphite crystal. After some time, these separate from the graphite flake by a collision-induced force and are dispersed in the NMP as exfoliated flakes. Approximately 0.034 mg mL^{-1} of graphene was produced after 2 min. Yield was increased to 0.11 mg mL^{-1} by prolonging the chaotic flow process to 30 min. Centrifugation was followed by redispersal of the top 80% supernatant in NMP. This solution was then sonicated. This two-step exfoliation process resulted in higher concentrations over time compared to direct sonication. A graphene sheet (GS) concentration of 0.430 mg mL^{-1} was reached after 8 h of sonication for the combined treatment compared to 0.119 mg mL^{-1} for pure sonication (8 h). These concentrations also appear typical for other sonication only studies, as seen in the previous section. Using AFM and TEM characterisation, a significant number of exfoliated flakes had a lateral size below $1 \mu\text{m}$ and in the range of several hundred

nanometers. AFM indicated that the thickness of exfoliated flakes was in the range of 0.5 and 1.8 nm, suggesting that the graphene sheets were ≤ 3 layers. Raman spectroscopy showed a slightly higher value of ≤ 5 layers thick. An advantage of this method is the reduction in processing time. One of the main drawbacks of this process, however, is the low yield ($<10\%$) of graphene from the initial graphite concentration.

Electrochemical exfoliation

The electrochemical exfoliation process utilizes the electrical conductance of the graphite precursor, to apply a positive or negative charge on the material. This mechanism is shown in Figure 11. Oppositely charged ions, intercalate the graphitic electrode and induce exfoliation. The most common intercalating species include halogens, alkali metals, metal halides, and various acids. In electrochemical exfoliation, the main components of the experimental set up include: a graphite working electrode, a counter electrode, a reference electrode, an electrolyte, and a power supply. Graphite flakes are normally adhered to carbon tapes to improve its conductivity as a working electrode. A typical experimental set up is shown in Figure 23a). The working electrode and counter electrode are immersed into an electrolyte with a certain distance kept between them. A positive (anodic electrochemical exfoliation) or negative (cathodic electrochemical exfoliation) voltage is applied to the graphite working electrode, depending on the desired exfoliation mechanism.

Liu et al.¹²³ utilized two pencil cores as graphite sources which acted as both the anode and the cathode. An alternating bias between $+7 \text{ V}$ and -7 V was used to exfoliate them. However, this led to inhomogeneity in the thickness and size distribution. Abdelkader et al.¹²² proposed a novel system to turn the process to a continuous operation as can be seen in Figure 23b). In this untested concept, graphite is inserted from the bottom, and areas in contact with the electrolyte would be exfoliated. Liu et al. combined the use of ultrasonication and electrochemical exfoliation simultaneously.⁵⁰ It was found that the selectivity of thinner layers of graphene was obtained by introducing sonication.

Anodic exfoliation

This method was initially used to produce acid salts of graphite since the 20th century. Interestingly, this substance

can form graphene flakes by applying high temperatures.¹²³ Parvez et al. used a series of inorganic salt aqueous electrolyte solutions in anodic exfoliation, and demonstrated that sulphate ions have the highest exfoliation energy. A yield of 75 wt % was achieved when the concentration of the anions was 1 M. The Raman D/G intensity ratio was about 0.25.²⁸ A mechanism was proposed for this process in which the production of hydroxide ions (i.e., strong nucleophile) occurs. These ions interact with the sp^2 carbons at the graphite edge, and further interactions with hydroxide ions result in the formation of an epoxide ring. This aids the expansion of the graphite layers, allowing the intercalating species such as sulphate ions, to promote exfoliation. The production of side gases, O_2 and CO_2 , has also been reported to play a role in the exfoliation process.

Many researchers have attempted to optimize this technique to mass-produce graphene efficiently using different electrolytes such as: sodium dodecyl benzene sulfonate as the electrolyte,¹²⁵ poly(sodium-4-styrenesulfonate) solution,⁴⁶ copper phthalocyanine tetrasulfonic acid,¹²⁶ 6-amino-4-hydroxy-2-naphthalene-sulfonic acid in a mixture of sodium hydroxide as the electrolyte.¹²⁷ These various routes have produced different types of graphene from mono-layer to MLG. The disadvantage of this method is that there may be oxidative functional groups that bound to the surface of the product, which may subsequently affect the final properties of the graphene obtained.

The advantage of high yield with low production times that anodic electrochemical exfoliation offers is strongly hindered by the production of graphene oxide. A lower oxygen content is, however, possible in comparison to normal rGO with the O/C atomic ratio (atomic oxygen to carbon ratio) ranging from 0.10 to 0.20.¹²⁷ Several researchers have tried to address this issue and a review by Paredes et al.¹²⁸ details some of the approaches that have been taken.

By tailoring the electrolyte or introducing additives to the solution, it has been shown by various researchers that the O/C atomic ratio can be minimized whilst maintaining efficient exfoliation. It has been found that the key issue with the process arises from the formation of oxygen radicals (e.g., OH).¹²⁸ Most of the strategies adopted, therefore, have aimed to minimize the effect of such radicals in the intercalation process. The use of an equimolar proportion of glycine and H_2SO_4 was investigated by Rao et al.¹²⁹ and it was found that the O/C atomic ratio was 0.12. It was further improved by Chen et al.¹³⁰ using melamine instead and obtained O/C atomic ratio of 0.04. Hossain and Wang¹³¹ investigated the use of H_2O_2 in the electrolyte and reported low oxidation. Reducing agents such as 2,2,6,6-tetramethylpiperin-1-yl oxyl (TEMPO) was used successfully by Yang et al.¹³² to get an O/C atomic ratio of 0.04 and I_D/I_G ratio of 0.1.

Cathodic exfoliation

In anodic exfoliation, the electrochemical exfoliation potential is greater than the oxidative potential. Hence, the graphene formed usually has oxidative species attached to it which affects its electronic structure. To circumvent this issue, researchers have tried to exfoliate graphene using it as the cathode electrode instead which reduces the extensive oxidation of graphene resulting in a structurally better product. Research in this field came from past applications of graphite when it was used as the negative electrode in lithium ion battery technology. Lithium can act as an intercalation agent and

form lithium hydroxide, freeing graphene sheets when the graphite-lithium intercalated compound decomposes in water.¹²⁴ A disadvantage with this technique is the slow kinetics, and many researchers have tried to optimize the process to overcome this problem.

To improve the kinetics of the process, Huang et al. used molten $LiOH/LiCl$ at $600^\circ C$, but failed to achieve complete exfoliation of graphite and needed sonication steps to achieve sensible yields.²² Zhong and Swager¹³⁴ proposed using Li^+ (from lithium perchlorate electrolyte) followed by tetraalkylammonium in two steps, however, a technical issue arising from the high ohmic drop created by the electrolyte, required high potential and still needed sonication treatment afterwards. Abdelkader et al. proposed a method using DMSO due to its intrinsic property of having a wide electrochemical window, while being an effective solvent to form graphene. The DMSO electrolyte contained triethylammonium and Li ions to produce large flakes of about $20\ \mu m$ lateral size. This setup also had the additional benefit of producing SO_2 /ammine gases that apply enough stress via expansion between the sheets to overcome the van der Waals attractions.¹²² Yang et al. used *N*-butyl, methylpyrrolidinium bis(trifluoromethylsulfonyl)-imide and claimed to get graphene sheets between 2 and 5 layers that were free of defects.³¹ Yang et al. also claimed to get pure cathodic exfoliation that did not require additional steps after using *N*-butyl, methylpyrrolidinium bis(trifluoromethylsulfonyl)imide ionic liquid but required potentials in the excess of $-30\ V$.³¹

Cooper et al. reported promising results using this technique. A tetraalkylammonium salt was used in NMP as the electrolyte, and produced purely cathodically exfoliated material with a low voltage of $-2\ V$ for approximately 48 h. The resultant graphene consisted of mainly 2–5 layers (thickness of 1.8 nm), exhibited some defects as indicated by the I_D/I_G of 0.64, and did not contain any signs of oxidation.¹⁵

Shear-assisted electrochemical exfoliation

Although electrochemical methods can exfoliate graphite into graphene, many issues arise from this technique. In the case of anodic exfoliation, the oxidation of the graphene has a significant effect on the quality produced. The slow kinetics in cathodic exfoliation limits its scalability in industry. Shinde et al. proposed a combination of two mechanisms to efficiently exfoliate graphite.⁴⁹ Applying minimal anodic potential, the oxidative effect was reduced. Simultaneously applying shear promoted the graphite expansion with intercalation of sulphate ions between the individual graphene sheets. A schematic of the micro-reactor in this experiment can be seen in Figure 24a. The effect of applying various levels of shear with electrochemical exfoliation was investigated and the outcome of this is presented in Figure 24b). Above 4 V, the process is electrochemically dominated. Below that voltage, the effect of the coupled interaction of the two mechanisms is evident. At the lowest voltage potentials (1–2V), the process is shear dominated. Under the applied shear, an initial expansion of the graphene sheet due to the intercalated molecules could have enhanced exfoliation. Mono- and bi-layer graphene was produced using this method and confirmed with AFM measurements. The potential applied played a pivotal role in determining the selectivity of the number of layers. At a low potential of 1 V, the average thickness of the graphene sheet was below 4 layers (less than 75%). In contrast, applying a 5 V potential resulted in thicker layers (5–6). Shinde et al.

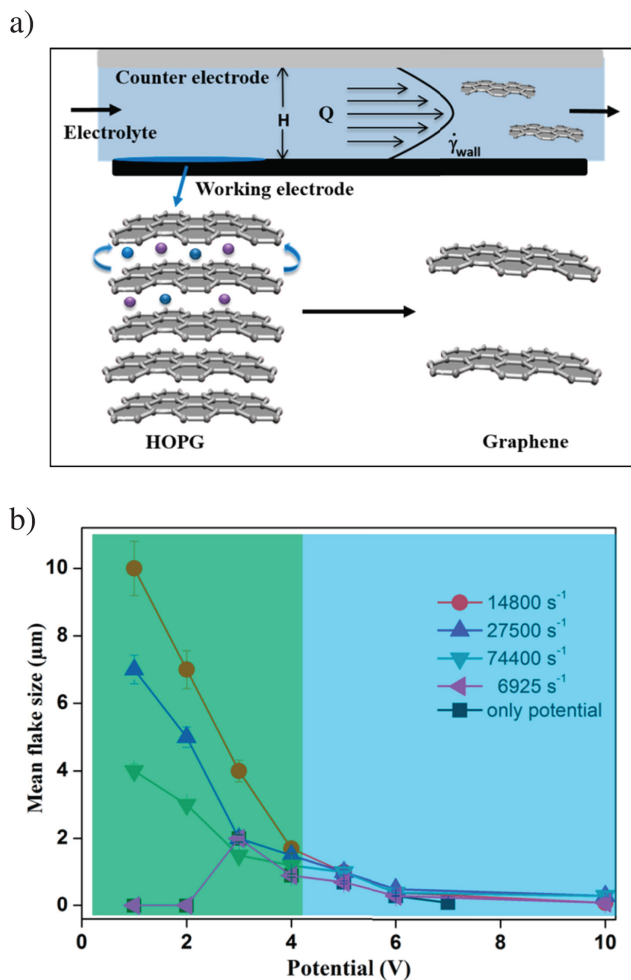


Figure 24. a) Schematic representation of the electrochemical microreactor combining electrochemical exfoliation and shear-driven exfoliation. The graphite crystal is both the wall and the working electrode of the reactor and simultaneously experiences a high wall shear rate and an applied electric potential. Here H , Q , and $\dot{\gamma}$ are the height between two electrodes, electrolyte flow, and shear rate, respectively⁴⁹; b) Effect of potential and shear rate on the size of graphene flakes produced in the flow reactor. A mean size distribution of zero, specifically in the case for only potential and shear rate of 6925 s⁻¹, indicates that exfoliation was unnoticed over the samples that were measured.

The green and blue colors in the figure indicate shear and potential dominated regions, respectively. Statistical flake size analysis for the graphene was performed using more than 80 sheets from AFM measurements. Error bars in these figures are from three samples at a given condition, showing the maximum and minimum values. Reproduced with permission from Ref. 49. Copyright 2016 American Chemical Society. [Color figure can be viewed at wileyonlinelibrary.com]

noticed that increasing the potential from 1 to 10 V increases the I_D/I_G ratio from 0.1 to 0.8. This suggests that oxidation of graphene occurred.

Challenges and Outlook

The previous sections highlight the range and diversity of the current methods within the domain of top-down, liquid-phase exfoliated production. As research surrounding the application of graphene and other 2-D materials has developed, two roadmaps have been generated which provide indicative timelines for production advancements.^{10,13} Novoselov et al.¹³ suggested in 2012 that for the following decade, the application of graphene would be driven by progress in production approaches. As evidenced in the current review, both top-down and bottom-up processes have expanded since 2012, but further improvement is necessary. There appears to be a disconnect between the proof-of-concept devices that are constructed on a lab-scale and the more effective approaches emerging as mass-production candidates. Dry mechanical exfoliation, using scotch tape, has been the primary method used at the lab-scale for studying new devices and fundamental physics. While this has resulted in impactful breakthroughs showing graphene's potential, it is a technique which is impractical for large-scale production. As material properties are directly linked to production method, it is likely that this disconnect contributes a certain amount towards the success rate and pace at which graphene-enabled technologies transfer into practical applications.

The scientific and technological objectives set out within the framework of the European Graphene Flagship¹⁰ for large-scale production remain the development of *reliable*, *reproducible*, *sustainable*, and *safe* approaches that can meet the needs of application areas. These key characteristics have a clear benefit for implementation, however, the previous literature on production approaches are predominantly lacking in these areas.

Control and characterization of material properties

There is limited information provided on the statistical methods used, if any, to ensure that the yield, layer distribution, lateral dimensions, and defects mentioned are repeatable or a true representation of the production approaches large-scale capability. Material characterisation techniques (AFM, TEM, SEM, etc.) predominantly deal with miniscule sample sizes and so this is challenging. Nonetheless, it is an area that should be addressed to achieve reliable and reproducible outputs. It would also ensure that comparisons between published data for emerging production approaches can be conducted without a concern for underlying measurement bias. This limitation is, however, beginning to receive attention. Recently, Liscio et al.¹³⁴ presented a thorough statistical analysis on fragmentation and size distributions due to ultrasonic exfoliation. Over 2500 GO sheets were analysed to determine sheet characteristics with confidence. Further effort is needed, both in terms of producing reliable insights on fundamental exfoliation mechanisms, and to develop standard protocols in industry.

The primary focus for the last decade, particularly for top-down production, has been the search for methods which can produce high yield and industrial scale production rates. Properties such as carrier mobility vary by up to six orders of magnitude, depending on the production approach. For the present moment, at least, coverage of a broad set of applications (i.e., from high-frequency transistors to conductive inks) will require a broad range of production approaches. However, in addition to high yield and production rates, demonstrating the ability to tune material properties within individual approaches

would also be beneficial.¹⁰ Research on tuning material properties by varying the operational parameters of a single production approach are currently limited. Paton et al.³⁵ correlated the effect of rotor speed on the resultant mean flake length for a high-shear mixing method. Liu et al.³⁶ used the same exfoliation method, and tested two different rotor sizes, to reveal the variations in flake length and thickness that can occur. Chen et al.¹⁰⁸ demonstrated with a spinning disc reactor that both full and partial carbon nanoscrolls can be created, depending on the morphology of the graphite precursor and the varying local microscale fluid mechanics. These liquid-phase exfoliation examples provide evidence that morphological properties of a 2-D material can be altered. By refining this ability to construct different material characteristics within a single approach, a broader variety of end-use applications can be targeted.

Solvents

Liquid-phase exfoliation methods require solvents to assist in production of graphene and form a stable dispersion. Suitable solvents are also required for storing graphene produced via other top-down methods. As it was demonstrated, a good solvent, or a mixture of few solvents, can be selected on the basis of their technical characteristics (see section on Solvents).⁵⁴ Ionic surfactants can also be added to poor solvents such as water to effectively assist in exfoliation process and prevent graphene from restacking. Unfortunately, most of the good solvents, surfactants and other relevant chemicals suffer from several issues for any potential scale-up technology as well as post-processing and storage. For instance, the best solvents currently used, such as NMP, have a high boiling point making them difficult to separate (and preferably recycle) from the final product without affecting the quality of graphene. Moreover, such solvents are toxic and can be incompatible with common materials of construction used in the plant. Thus, when one considers recyclability, health and safety, and environmental aspects of their process, these issues can introduce additional constraints from an engineering and financial perspective.

The unavailability of a suitable solvent that could be used for large-scale production will also cap the throughput per each production line (batch or continuous). From a storage and logistics point of view, it is highly desirable to maximize the concentration of graphene in a solvent and to maintain its initial product specification for as long as possible. A recent study has presented promising results in addressing the aforementioned solvent challenges.¹³⁵ Using a selection criteria that matched a range of solvents to a set of ideal dispersion characteristics that include polarity, toxicity, surface tension, viscosity and greenness, Cyrene was discovered to be the most suitable high-performance solvent for sustainable production of graphene. This solvent is environmentally benign, and in this work, facilitated graphene concentrations an order of magnitude greater than NMP.

Conversely, graphene is a lightweight material and occupies a much smaller volume in solid phase than when dispersed in solvent, resulting in far more economical transportation and storage. So there is also a scope for development of a process that can efficiently separate graphene from its solvent without negatively affecting any of its superior qualities, or encouraging aggregation. Therefore, to counteract all of the aforementioned issues, it is probable that there is going to be a need for a synthesis of specific solvents, surfactants or other chemical

compounds that possess targeted properties. Furthermore, a unique pathway may also need to be designed that will preprocess and postprocess graphite/graphene dispersion using an efficient method. It is crucial to design a product which could be effectively utilized by various downstream industries.

Rheology

The challenges in engineering environmentally friendly, low boiling point solvents for the dispersion of graphene are nontrivial. High-boiling point solvents, with toxic characteristics, currently produce the highest graphene concentrations. A growing body of empirical and numerical research demonstrate that the long-term stability of monolayer number fraction is limited to tens of days, even for these high performing solvents. Solvent design may be further complicated if the end-use of liquid-phase exfoliated graphene involves solution processing. In the rapidly emerging area of printable graphene inks for flexible and conductive electronics, rheological properties that include surface tension, viscosity and density impact the printing process.³⁴ NMP has properties that are suitable for implementation in ink-jet printing, however, both the toxicity and post-treatment requirements (removal at >170°C) complicate and limit its use. Aqueous surfactant solutions are also limited in practical applications, as the stabilizing agents have an adverse effect on electronic properties. Capasso et al.³⁴ recently tuned the properties of water-ethanol (H₂O/EtOH) mixtures for the exfoliation, dispersion and formulation of graphene-based inks. A 1:1 volume ratio provided a surface tension of 30.9 mN/m, which was within the range required for the ink-jet printing process. The transmittance and conductivity were compared for H₂O/EtOH films and NMP films, both of which had no post-processing applied. A promising outcome was that H₂O/EtOH films could provide higher transmittance at lower electrical resistances by increasing the number of low concentration printing passes. Developing solvents and co-solvent strategies, with tuneable rheology for the end-use, will provide a significant advancement towards introducing this material into practical applications.

Integration with chemical engineering processes. The growing research field on scale-up of graphene is at a relatively early stage. Most of the previous discussion has surrounded the enhancement of yield, production rate and quality. While this is fundamentally crucial for advancing production methods, there are wider engineering process challenges to be considered. The successful practical implementation of the most promising top-down methods will also rely on seamless system-level integration. Figure 25 provides a summary of the key aspects involved in an end-to-end production process (Steps 1–6). At present, some components are more developed (e.g., precursor choice, solvent and aqueous surfactant solutions, and exfoliation methods) than others (e.g., high-throughput graphene separation, quality assessment, recycling, storage/handling). This illustration also highlights the dependency of exfoliation methods (step 4) on factors along the process chain. Robust technologies, capable of easily integrating within this process chain, have a distinct advantage. A holistic view of the entire chemical engineering process, therefore, is required for graphene production to translate from the laboratory to industrial scales.

Modeling and simulation

The liquid-phase exfoliation process is complex, covering a range of length scales that span 6–9 orders of magnitude, from the device level (10^{−3}–10⁰ metres) to the graphene end-product

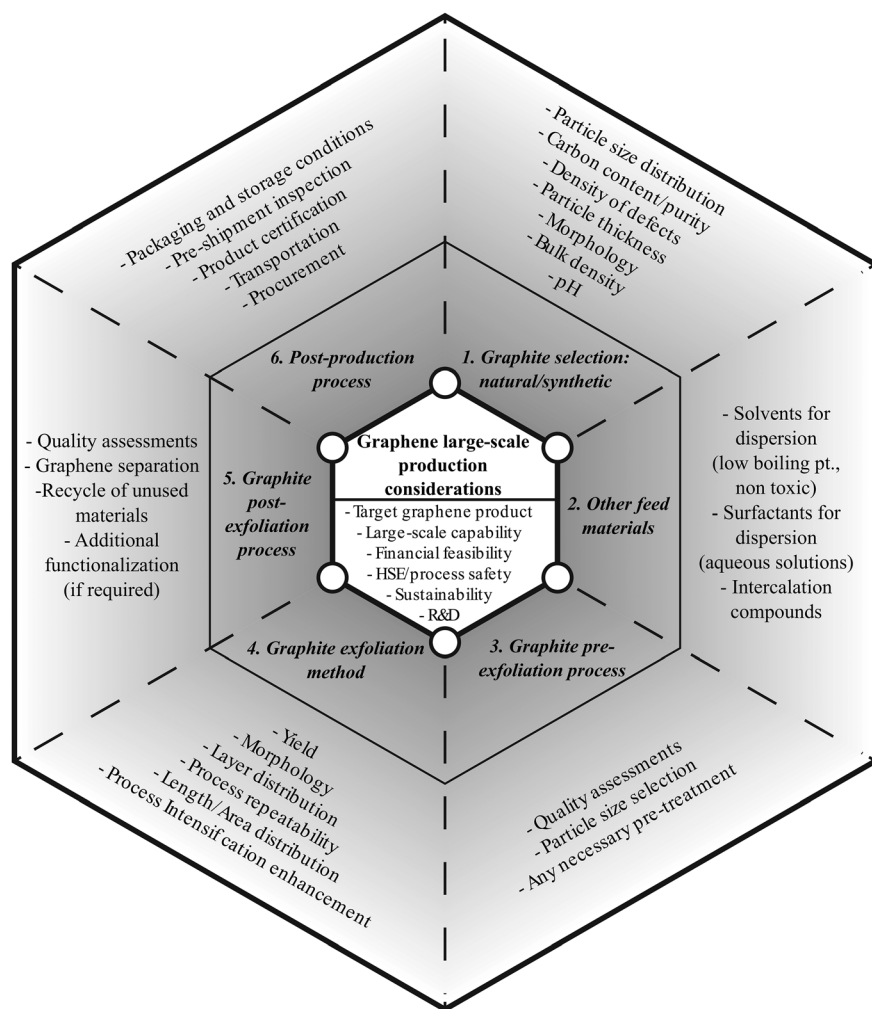


Figure 25. A summary of the top-down production considerations in the scale-up of graphene.

(nanometres). Processing time is typically over hours, whereas the individual exfoliation mechanisms can occur at fractions of a microsecond. Examples of this are in the approaches that utilize the rapid bubble collapse during cavitation¹³⁶ to induce exfoliation of graphite. In addition to a discrete phase (graphite), multiple continuous phases (liquid and gas) exist in shear-driven methods such as sonication, jet cavitation, and thin film approaches like the spinning disc and vortex fluidic devices. Multiple species (intercalants, oxidising agents) exist in chemical and electrochemical methods, and the microwave exfoliation method is an example of a multiphysics phenomenon. The breadth of transport phenomena and scales is, therefore, significant. This makes it challenging to analyse the exfoliation processes in high fidelity through experiments alone.

Numerical modeling is a powerful tool for providing fundamental insights that can be validated with experiments. Molecular dynamics has already been used to understand the interactions between graphene sheets, when dispersed in various solvents⁶⁷ and aqueous surfactant solutions⁶⁹ (see section on Graphene Dispersion). Chen et al.¹⁰⁸ observed carbon nanoscrolls in exfoliation experiments, and utilized molecular dynamics to study the conditions that favor such scrolling behavior. Other techniques, such as computational fluid dynamics (CFD), can elucidate fluid and transport phenomena at larger scales where a continuum representation is valid. These modeling and simulation tools are well-established in

chemical engineering disciplines, but have yet to be utilized to their full potential for liquid-phase exfoliation. As this research field continues to develop, it is anticipated that these tools will also be important in practice, for the prediction and optimisation of production performance.

Other 2-D materials

In addition to the improvements necessary for graphene production, other 2-D material research has opened exciting new avenues for large-scale production. Similar to graphite, transition metal dichalcogenides (TMDs) and transition metal oxides (TMOs) also have a layered structure with van der Waals forces holding individual layers together. There is potential to discover many other layered materials with novel properties that have yet to be isolated into 2-D form. These materials can be realized over the next 5 years on a large-scale by extending proven production technologies for graphene to other 2-D crystals. An example of this is the recent work by Cullen et al.¹³⁷ on dissolution of numerous 2-D materials (Graphene, TiS₂, MoSe₂, Bi₂Te₃) to form ionic liquids.

Long-term research challenges beyond the next decade include the production of 2-D heterostructures, or stacked layers of different materials which are held together through van der Waals forces. These materials have shown uniquely beneficial properties at the lab-scale, such as the highest-mobility graphene transistors, created by encapsulating

Table 4. Summary of the Bottom-Up and Top-Down Production Methods, Highlighting Key Graphene Characteristics as well as the Advantages and Disadvantages of the Various Approaches

Method	Advantages	Disadvantages	Process Details	Raman D/G ^a	Selectivity	Yield (wt %)	Production rate (g/h) ^b	Reference
Chemical Vapor Deposition (CVD)	Potentially high production rate	Defective graphene Harsh conditions	From sodium ethoxide	1 (532 nm)	Higher selectivity to 4 layers	–	–	141
Arc Discharge	High quantity of graphene can be produced Low processing time required	Defective graphene	Atmospheric Pressure CVD	–	–	–	–	142
			Cold-wall CVD	0.2	–	–	–	143
			High speed roll to roll	0.16	–	–	–	144
			In Hydrogen In N ₂ and H ₂ with AC current	– 0.55–0.92 (633 nm)	2–4 layers Mainly 2 layers	10–20 5.3	– 25.2	145 30
Epitaxial growth	High coverage of monolayer across the metal surface Low cost	Defective graphene	Helium with O ₂ and H ₂	~0.8 (514.5 nm)	6 layers	–	–	146
			H ₂ -inert gas mixture	0.26–0.31	Mostly 2–4 layers with a small portion of 5–10 layers	–	~3	147
			On SiC	–	1–2 layers	–	–	148
			In pyrene surfactant In NMP	0.33 (633 nm) Very low D band observed	~28% monolayer. <5 layers	– 8.3–12	0.02 8×10^{-3}	75 21
Sonication	Low concentration of graphene	Defective graphene Low concentration of graphene	In surfactant	0.93 (514 nm)	–	–	0.018	149
			In isopropanol	0.2–0.4 (633 nm)	–	–	0.015	150
			In NMP	0.15 (514nm)	–	–	–	151
			In chloroform In surfactant, NMP and <i>ortho</i> -dichlorobenzene (o-DCM)	– –	– ~28% monolayer, 19% bilayer.	– 1.2	8×10^{-3} –	152 153
Solvothermal	Selective to monolayer graphene High yield	Long processing time Harsh conditions High pressure	In gum Arabic	0.25 (633 nm)	Mainly 5–10 layers	5	5×10^{-3}	154
			In nonionic polymeric surfactant	0.35	Mainly 2–3 layers	–	–	155
			In surfactants In NMP	– 0.2–0.5	Mainly 4 layers 90% less than 5 layers	0.5 4	5×10^{-3} $\sim 5 \times 10^{-3}$	29 23
			In SDBS surfactant	0.4	3% monolayer, 40% <5 layers	3	–	40
Supercritical Fluid	High yield	Difficult solvent/graphene separation	Two-step in NMP	<0.5 (633 nm)	3–4 layers	19	7.5×10^{-3}	24
			In forming gas	–	1–2 layers	10–12	8.3×10^{-5}	16
			In tartaric acid	0.06	~95% monolayer	–	–	96
			In sulphate dispersant In NMP, DMF	– –	10 layers 3–10 layers mostly	40 10	0.8 3.3×10^{-4}	43 104
Microwave	High yield Short processing time Simple and implemented easily technology Simple setup with minimal issues during the operation	Very low yield	Shear assisted Ultrasonication assisted	0.56 –	~70% 5–8 layers 24% 1 layer, 44% 2 layers, 26% tri layers	39 21.5	0.9 0.11	39 20
			In Ionic liquid	0.14	95% monolayer	93	0.02	27
Vortex fluidic	High yield Short processing time Simple and implemented easily technology Simple setup with minimal issues during the operation	Low yield Low yield	In NMP	–	–	1	–	106
Taylor-Couette Pressure Driven	High quality graphene Simple process High quality graphene	Low yield Low yield	In NMP Jet Cavitation	0.14 0.38	90% less than 5 layers 70% less than 5 layers	5 5	– 0.25	47 26

TABLE 4. Continued

Method	Advantages	Disadvantages	Process Details	Raman D/G ^a	Selectivity	Yield (wt %)	Production rate (g/h) ^b	Reference
High shear mixing	High production rate High quality graphene	Precise control of operating conditions required Defective graphene Expensive Long processing time	In IPA-water	0.14, 0.18 (514nm)	Mainly less than 4 layers	8	—	36
			Mixer	0.17–0.37	5–8 layers	—	5.3	35
			Kitchen blender	0.12	—	7.3	9.2×10^{-3}	110
Micromechanical cleavage	High quality graphene Selective to monolayer	Defective graphene Expensive Long processing time	With scotch tape	—	monolayer	—	—	2
			With wedge	0.86	—	50	—	156
Ball milling	Cost effective Potential for graphene functionalization	Very long processing time Defective graphene	With three roll mill	0.06	50% monolayer	90	Max-0.075	19
			In SDS	0.6–0.7 (532nm)	—	50–75	1.5–2.5	25
			In DMF	0.34	1–3 layers	—	—	157
			In methanol and 1-pyrene carboxylic acid	0.24–0.38 (488nm)	5–10 layers	—	—	158
			In sulphur	—	95% <10 layers	—	—	159
Spinning disc	Possible continuous flow configuration Cost effective	Long recycling time	In gases	1.32	25% monolayer	—	—	160
			In NMP	—	Most less than 5 layers	1	—	108
Chemical exfoliation	High yield	Low production rate Defective graphene	By acetic acid	1.1	—	—	0.06	161
			By Na in NH ₃	1.4	—	—	0.33	162
			Staudenmaier	—	—	—	1×10^{-3}	163
Electrochemical exfoliation	High yield	Expensive Defective graphene Limited number of studies reported	Hummer method	—	85% less than 3 layers	—	0.45	164
			Anodic exfoliation	0.25	72% 1–2 layers	75	32.6	28
			Anodic with Poly(sodium-4-styrenesulfonate) PSS	—	2–5 layers	15	—	46
			Anodic with copper acid	0.92	1–6 layers	—	—	126
			Anodic with sulfonic acid	0.32	5% 1–2 layers 78% 4 layers	—	—	127
Chaotic flow with sonication	Short processing time	Low yield	Cathodic with LiOH	0.29	2–4 layers	80	1.1	22
			Cathodic with Li ⁺	1.1	—	40	—	133
			Cathodic with DMSO	0.5	5% monolayer	—	—	122
			Cathodic with ionic liquids	0.05	2–5 layers	25	5.56×10^{-3}	31
			Anodic with shear assisted	0.1	75% less than 4 layers	—	—	49
			In NMP	—	Less than 3 layers	7.2	—	38

^aThe wavelength in parenthesis indicates the wavelength of the laser used for Raman spectroscopy analyses.^bProduction rate was estimated from best possible yield (mass of graphene/the mass of the initial graphite precursor) and processing times from the paper.

graphene with h-BN using CVD.¹³⁸ Flexible photovoltaics have also been demonstrated utilizing optically active semiconducting layers (TMDs) with graphene as transparent electrodes.¹³⁹ Novoselov et al.¹⁴⁰ noted that this topic is now at a stage where graphene was 10 years ago, with “*plenty of interesting science and unclear prospects for mass-production.*” If research efforts grow at a similar rate to what has been realized for graphene over the past decade, there is an exciting future for both the scientific and application areas of 2-D material production.

Conclusions

The topic of large-scale production has grown considerably over the last decade to establish itself as a scientific field within the overall graphene research domain. Developments have evolved into process routes that can now be categorized under two independent production approaches. These approaches, defined as “top–down” and “bottom–up,” produce a variety of graphene-based materials, including monolayer and multilayer sheets, nanoscrolls, and graphene oxides. Liquid-phase exfoliation is a top–down method which has a high potential for scale-up. This approach has demonstrated the highest production rates ($\sim 6 \text{ g h}^{-1}$), but with comparatively low yield ($<5\%$). Electrochemical exfoliation achieved high yield graphene oxides in studies using HOPG ($>70\%$), and when combined with shear-assisted flows this method has produced low-defect flakes, which have some of the largest dimensions ($\sim 10 \mu\text{m}$) for top–down approaches.

The notable disparity in product characteristics across top–down approaches is linked to the multiple exfoliation mechanisms that have been investigated. Shear, collision, cavitation, chemical, and electrochemical intercalation, the type of graphite precursor, solvent selection, and the inclusion of postprocessing techniques such as centrifugation all impact graphene output. In contrast, bottom–up methods such as CVD and epitaxial growth, are less diverse and have the capability to synthesize high-quality monolayer sheets with larger lateral sizes ranging from tens of μm^2 to several cm^2 . While this is preferable for applications that require high-carrier mobility ($\sim 10^5 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$), there is no clear indication of potential rates of production. Bottom–up methods can also require postprocessing steps to remove the graphene sheets from the substrate on which it has been grown. The lack of convergence towards a single all-encompassing method is due to the material requirements of the potential end-use applications. Energy storage and generation, flexible and printed electronics, and composites can be addressed using top–down approaches. High-end electronics and the ability to integrate with semiconductor devices are applications which need a high-quality product achievable by bottom–up approaches.

Across all production approaches, it is anticipated that the future developments for graphene will be around tunable morphology, proven reliability and repeatability, and sustainable and safe implementation in chemical engineering processes. This will ensure that scientific outputs can be translated into commercial technologies that can meet application demand. Finally, a summary comparing the production characteristics of different methods can be seen in Table 4.

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Notation

2D	= two-dimensional
3D	= three-dimensional
1D	= one-dimensional
0D	= zero-dimensional
BNNS	= boron nitride nanosheets
TMDs	= transition metal dichalcogenides
TMOs	= transition metal oxides
TEM	= transmission electron microscopy
MLG	= multilayer graphene (also known as few-layer)
N_i	= layer numbers
GO	= graphene oxide
rGO	= reduced graphene oxide
SEM	= scanning electron microscopy
HRTEM	= high resolution transmission electron microscopy
AFM	= atomic force microscopy
SWCNT	= single-walled carbon nanotube
I_{2D}	= Raman 2D peak
I_D	= Raman D peak
I_G	= Raman G peak
XPS	= X-ray photoelectron spectroscopy
SAED	= selected area electron diffraction
h-BN	= hexagonal boron nitride
$\dot{\gamma}$	= shear rate
CPO	= cyclopentanone
$\chi \sim R^2$	= Flory-Huggins parameter
MD	= Molecular dynamics
PMF	= the potential of mean force
SDBS	= Sodium dodecylbenzene sulfonate
SC	= sodium cholate
GIC	= graphite intercalated compound
PGO	= pristine graphite oxide
ρ	= liquid density
η	= liquid viscosity
GS	= graphene sheet
CG	= concentration of graphene (supernatant)
HOPG	= Highly Oriented Pyrolytic Graphite
IP	= intellectual property

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